

27 - 28 August
2025
VICTORIA FALLS

ATLAS CONVENTION

GLOBAL ACADEMIC CONFERENCE

PROGRAM &
BOOK OF ABSTRACTS



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Dr. Teurai Matekenya – Programme Chair

Dr. Inyene Nkanta – Organising Committee Member

Prof. King Omeihe – Organising Committee Member

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PROGRAM

IST ATLAS GLOBAL ACADEMIC CONFERENCE 2025

27 – 28 August

**Bridging academia, industry and
commerce for inclusive growth**

Day I: Wednesday 27 August 2025 Venue: Elephant Hills, Victoria Falls

TIME	ACTIVITY	FACILITATOR
08.00-08.30	Arrival and registration	Secretariat
08.35-08.50	Welcome Remarks & Overview of the 1 st Edition of the 2025 Atlas Global Academic Conference by Ephias Ruhode	Teurai Matekenya
	Introduction of the Day 1 Keynote Speaker	
08.50-09.20	Keynote Address: Prof. Rukuni. [Board Chair Zimbabwe Council of Higher Education]	Ephias Ruhode
	BREAK AWAY SESSION A	PRESENTER/AUTHORS
THEME: Regional and Transnational innovations in Agriculture and Food security:		
SESSION CHAIR: Lister Matsilele		
09.25-09.40	Transforming Agrotourism in Africa: Harnessing Digital and Social Media Marketing Strategies for Overcoming Business Challenges Among Zimbabwean Entrepreneurs	Theo Tsokota, Doreen Nkala; Chipu Mutongi
09.45-10.00	When Less Becomes More: Frugal Innovation in Micro-Farming for Climate-Resilient and Food Systems in Northern Nigeria	Inyene Nkata; Abdulrasaq Ajadi Ishola; Tokunbo Alaga Olorundami
10.15-10.30	Strengthening Farmer's Adaptation to Climate Change Impacts on Pest Management, Weather Forecasting and Crop Planning: How Appropriate are Digital Tools?	Prof Ngoepe-Ntsoane, Sifiso Fana Mdluli; Nonkululeko Ndaba; Funeka Manzi
10.35-11.00	HEALTH BREAK	
Theme: Growing the rural economy and rural industrialization:		
Session Chair: Nobert Jere		
11.00-11.15	Effectuation and Bricolage and their Applicability to Sub-Saharan African Entrepreneurship	Prof Michel Rod
11.20-11.35	Innovative Agro-Processing Technologies as Catalysts for Rural Industrialization and Economic Growth. "A case study for Marange in Mutare.	Juliet Mupete; Kupara Sencilia

11.40-11.55	Colonialism's enduring legacy: Unpacking the Roots of rural underdevelopment, curving a path for growing the rural economy in Zimbabwe-The case of Nembudziya Gokwe North	Tapiwa Musasa
12.00-12.15	The impact of rural industrialisation on community health and well-being in Zimbabwe.	Plaxedes Makoni; Njabulo Nkomazana
12.20-12.35	Sustainable financing rural economic development in southern African countries. A systematic review	Shadreck Nhorito; Prof Collins C Ngwake
12.40-12.55	Growing rural economies leveraging beekeeping: Income implications of beekeeping value chain integration.	Amon Taruvinga, O Zamisa
13.00-13.15	Navigating the effects of load shedding on agricultural productivity and food security: A case study of South Africa,	Prof Ngoepe-Ntsoane
13.20-14.00	LUNCH BREAK	
	BREAK AWAY SESSION B	
Theme: Societal implications of technology, smart cities and smart villages in Africa:		
Session Chair: Amon Taruvinga		
09.25-09.40	The Role of Entrepreneurs in Advancing Smart Cities and Smart Villages in Zimbabwe: Societal Implications of Technology	Doreen Nkala, Chipso Mutongi; Theo Tsokota
09.45-10.00	Societal Implications of Technology, Smart Cities, and Smart Villages in Africa	Nontethelelo Tshivhase; Sanderson Abel
10.15-10.30	Technoethics and Heuristic Reasoning: Ethical Heuristics for Advanced Technology Adoption in South African Digital Transformation Initiatives	Amanda Vester, Prof Eric Schaling, Prof Theo Haupt, Prof Ephias Ruhode
10.35-11.00	HEALTH BREAK	
Theme: Tech-Driven Futures: Advancing Work, Education, and Health in Africa.		
Session Chair: Christinah Moyo		
11.00-11.15	Challenges-faced-in-implementation-of-green-procurement	Melisa Ncube; C Sinchansambwe; Sultan Al Kaabi SQ
11.20-11.35	An Improved Linac Mechanical Quality Assurance Device (The LINATHAD) and its Implications in The Medical Field.	Prince Chitopho
11.40-11.55	The Future of Work: Navigating the intersection of automation, Remote Work, and Human-Centred Skills in the digital revolution	Herbert Masukume
12.00-12.15	Skills, Education, Employment, and Sustainability in Renewable Energy SMEs: A Namibia-Zimbabwe Cross-Country Comparison	Marcus Mutanga; Venaan Paulus; Enard Mutenheri, Mbela Kalengay
12.20-12.35	The efficacy of technical vocational education and training programs in aligning graduate skills with the demands of Africa's emerging green economy: A literature review.	Wilson Mabhandanda
12.40-12.55	The Role of Intellectual Property in Driving the Operationalization of Education 5.0 in TVET Institutions: The Case of Kwekwe Polytechnic, Zimbabwe	Sylvia Tendai Moto; P. Muredzi
13.00-13.15	Climate Change and Pest Management: A Systematic Review of Digital Tools, Integrated Strategies, and Climate-Driven Pest Outbreaks	Nonkululeko Ndaba
13.20-14.00	LUNCH BREAK	

14.00-15.30	Panel Discussion: AFRICA FORUM Prof Mandy Rukuni 1. Prof Gabbey Malope [Online] 2. Prof King Omeihe 3. Prof Ngoepe-Ntsoane	Moderator: Ephias Ruhode
15.35-15.50	X-Culture	Ms Agnes Nsor
16.00 -	Excursion	
END OF DAY 1		

Day 2 Thursday 28 August 2025 Venue: Elephant Hills, Victoria Falls

TIME	ACTIVITY	FACILITATOR
08.00-08.05	Welcome & Introduction of the Day 2 Keynote Speaker	Teurai Matekenya
08.10-08.30	Keynote Presentation: Dr Electdom Matandirotya, Centre for Education Innovation Research and Development, University of Zimbabwe	Teurai Matekenya
BREAK AWAY SESSION A		PRESENTER
Theme: Partnerships for Prosperity: Fintech and Inclusive Growth in Africa:		
Session Chair S Nhorito		
08.35-08.50	Fostering manufacturing entrepreneurship through polytechnic-industry collaborations: A pathway to economic growth in Zimbabwe	Christinah Moyo; Theresa Mberikwazvo
08.55-09.10	Public-Private Partnership and Indigenous Knowledge Production for African Development	Bamigbose
09.15-09.30	Framework for Cryptocurrency Adoption as a Digital Currency by Zimbabwean Companies	T. Tsokota; S. Mandizvidza; C T Chifumbu; Kangare; C Mutongi
09.35-09.50	Impact of Bank-Fintech Collaboration on Financial Inclusion in Sub Saharan Africa.	Manford Moyo; T M Mango
09.55-10.10	Business Intelligence Framework on electronic banking in Zimbabwe Financial Service Sector.	Effort Mtombeni
10.15-10.30	Financial Inclusion of Rural Young Women In Zimbabwe: A Case of Saving Clubs in Ward 2 in Chiredzi.	S C Matema; T Chiparausha; G Musvipa
10.35-11.00	HEALTH BREAK	
Theme: Sustainable development and inclusive digital transformation:		
Session Chair: Theo Tsokota		
11.05-11.20	Explainable Artificial Intelligence in Healthcare: Applicability in African Communities	Jere Nobert
11.25-11.40	Leveraging Digital Marketing for Inclusive Growth: Sustainable Strategies to Bridge the Digital Divide in Zimbabwe.	Rukudzo Alison Mawere
11.45-12.00	Sustainable development and Inclusive Digital Transformation	P Marandure; Nyasha Zibwowa

12.05-12.20	Inclusive Digital Transformation for Sustainable Development to Bridge the Digital Divide in Sub-Saharan Africa: A Case Study in Zimbabwe	R Chigariro; Dr S Musungwini
12.25-12.40	Overcoming Obstacles and Maximizing Opportunities for AI-Driven Research in Zimbabwe's Indigenous Languages: Pathways to Innovation and Inclusive Sustainable Economic Growth	M Mazuruse; N Nhemachena
12.45-13.00	When Innovation Harms Inclusion: A Critical Analysis of Anti-Competitive Risks, Regulatory Gaps, and SME Marginalisation in South Africa's Emerging Drone Economy	Amanda Vester, Prof Eric Schaling, Prof Theo Haupt, Prof Ephias Ruhode
13.05-13.20	Zimbabwe through the lenses of Three African content creators.	Witness Roya
13.25-14.00	Lunch	
	BREAK AWAY SESSION B	PRESENTER
Theme: Digital Bridges to Global Markets: Modernizing Governance for Africa's Industrial Future:		
Session Chair: Slyvia Tendai Moto		
08.35-08.50	Assessing the attractiveness of African countries to Chinese Foreign Direct Investment using non-parametric approaches	Abson Chompolola.
08.55-09.10	The US-SA Media briefing as a theatre of global diplomacy in the digital era	Witness Roya
09.15-09.30	Sanctions, Protectionism, and Food: Assessing the Consequences for African Agricultural Trade and Food & Nutritional Security	Shandini Naidoo
09.35-09.50	Bridging the digital divide: An assessment of e-learning accessibility and impact on skill development for rural youth in Zimbabwe's secondary education sector	Wilson Mabhandha
09.55-10.10	Entrepreneurial creativity and small firm internationalisation: A theoretical nexus.	Benjamin Byarugaba
10.15-10.30	Bridging Borders, Building Futures: The Role of International Education in Shaping Africa's Workforce and Entrepreneurial Ecosystem	Agnes Nsor
10.35-11.00	HEALTH BREAK	
Theme: Corporate Governance and Social Responsibility.		
Session Chair: Herbert Masukume		
11.05-11.20	Strengthening community partnership for conservation: Gonarezhou Conservation Trust's engagement with Shangani people in Chiredzi, Zimbabwe	Lister Matsilele
11.25-11.40	Technology's Impact on Innovation in Education 5.0: Navigating Challenges of Intellectual Property in a Digital Age – Case of Kwekwe Polytechnic, Zimbabwe	Slyvia Tendai Moto; Margret Simango
11.45-12.00	Exploring the provision of cattle-related and other agricultural information using the South African Ilanga local newspaper, in KwaZulu-Natal Province of South Africa	Simphiwe Mngomezulu-Dube
12.05-12.20	The contribution of digital skills and knowledge to university students' inclination to embark on the fashion and beauty digital entrepreneurial start-ups in South Africa: a scoping review	Maphanga; Smangele Pretty Moyane; Ntando Nkomo; Collence T Chisita

12.25-12.40	Strengthening Climate Resilience through Digital Weather Forecasting: A Systematic Review with a Case Lens on Tomato Farmers in South Africa	Sifiso Mdluli
12.45-13.00	AI-Enabled Crop Planning for Smallholder Farmers: Enhancing Productivity and Climate Adaptation in South Africa	Funeka Manzi
13.05-13.20	Unpacking Policy Theory and Practice: Evaluating Service Provision and Uptake in Higher Education in Zimbabwe	Fungai Nora Mukora; Chadzimura CC; Kagondo BVT; and Nherera CM
13.25-13.30	Closing and Vote of Thanks	Ephias Ruhode

KEYNOTE SPEAKERS



PROFESSOR MANDIVAMBA RUKUNI

Professor Mandivamba Rukuni (Mandi) is a development analyst and strategist in several sectors including agriculture, land governance, rural industrialisation, and education. He is currently actively engaged with the Government of Zimbabwe, business sector and civil society on a number of developments including land tenure reform, rural industrialisation, reparations and fertiliser industry reforms.

He is also currently Chairman of Zimbabwe Council for Higher Education (ZIMCHE) and Chairman Great Zimbabwe University (GZU). An agricultural economist by training, Professor Rukuni taught for 20 years at the University of Zimbabwe and during his tenure served as Dean of Agriculture after which he worked for the W.K. Kellogg Foundation for 11 years, playing an instrumental role as their Director for Africa Programs.

He served on several corporate boards including: Kingdom Bank (Chair); Triangle Ltd; Zimsun Hotels; Zimnat Life; Blue Ribbon Industries, Hunyani Paper, and others. He served on boards of parastatals: ARDA (Vice Chair), GMB and Agricultural Research Council (Chair). Internationally he has served as Africa Director for W.K. Kellogg Foundation. Also currently serves as consultant and advisor to Africa Union, NEPAD, SADC, AfDB, World Bank, FAO, UNDP, and others.

The international boards he has served include: IFPRI; CIAT; Univ. of Sussex; and FAO. During 1993-4 he served as Chairman of the Presidential Commission of Inquiry into Land Tenure Systems of Zimbabwe. He was twice nominated (1990 and 1992) for Africa Prize for Leadership in Sustainable End of Hunger. He has published upward of 100 articles, several academic books; also, several non-fiction books including *Being Afrikan* and *Leading Afrika* (Penguin).



Dr ELECTDOM MATANDIROTYA

Dr. Electdom Matandirotya is a dedicated space science and space technology researcher who is always ready to meet daily challenges. Throughout her career, she has not only shown a commitment to achieving excellence but also her passion for outreach and mentorship.

She holds the following academic qualifications; a DEng in Electrical Engineering (Cape Peninsula University of Technology, South Africa, 2017 in collaboration with the South African Space Agency-SANSA) ; MSc in Electronic & Electrical Systems (Ecole Supérieure d'Ingénieurs en Electrotechnique et Electronique, France, 2013 in collaboration with the French South African Institute of Technology-F'SATI); MTech in Electrical Engineering (Cape Peninsula University of Technology, South Africa, 2012) and a BSc in Applied Physics (National University of Science and Technology, Zimbabwe, 2007).

Dr Matandirotya is currently the Acting Chief Scientist for the Centre for Education Innovation Research and Development (CEIRD) a centre that is dedicated to undertake research and innovation through fostering high-quality Education, Innovation, Research and Development in strategically important sectors that have impact on the economy and society of Zimbabwe. As a researcher she is affiliated with the Zimbabwe National Geospatial Space Agency (ZINGSA) and the Space Science and Applied Physics of the University of Zimbabwe.

The diverse roles that she has played in these institutions has empowered her with leadership, management and mentorship skills. As such she is part of several space science and space technology initiatives that the nation is currently implementing. With her research interests in Space Weather, Ionospheric modelling and Satellite Systems Design she has become a key partner in national and international collaboration initiatives that supports human skills capacity building, innovation and industrialisation.

CONFERENCE ABSTRACTS



NAVIGATING THE EFFECTS OF LOAD SHEDDING ON AGRICULTURAL PRODUCTIVITY AND FOOD SECURITY: A CASE STUDY OF SOUTH AFRICA,

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Eskom's electricity generation falls short of meeting the country's increasing demand, with an installed capacity of 52,000 MW, but actual production around 47,000 MW, creating a significant gap. To address this discrepancy and protect the national grid, Eskom has implemented voluntary power outages, known as load shedding. These outages have severely affected the agricultural sector and food security in South Africa. The continuous power cuts have disrupted farming operations, from irrigation to food processing, hindering farmers' ability to sustain production levels. This has resulted in crop losses, diminished livestock productivity, and delays in food distribution. Consequently, food prices have surged, exacerbating food insecurity, especially among vulnerable groups. A desktop review was conducted to analyse the load shedding impact and to evaluate the legislative framework that governs electricity supply and load shedding management in South Africa. The secondary data collected during the research helped clarify key issues, especially the policy and legislative gaps that may need to be addressed to ensure more effective management of the energy crisis. The study was conducted to examine the impact of load shedding on agricultural production and assessing the economic effects on food security. By focusing on agricultural productivity and its relationship with electricity supply, the study was able to propose evidence-based solutions and recommendations to help mitigate the negative impacts of load shedding on South Africa's food systems and ensure better food security for all.

Keywords: Load shedding, Agricultural productivity, Food security, Electricity supply, South Africa



THE ROLE OF ENTREPRENEURS IN ADVANCING SMART CITIES AND SMART VILLAGES IN ZIMBABWE: SOCIETAL IMPLICATIONS OF TECHNOLOGY

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This study investigates the societal implications of technology in the context of smart cities and smart villages in Africa, with a specific focus on Zimbabwe's growth points and the role of entrepreneurs. The rapid integration of technology into urban and rural settings presents both opportunities and challenges, necessitating a comprehensive understanding of its impact on social structures, economic development, and sustainability. Despite the potential benefits of smart technologies, many regions in Zimbabwe face significant barriers to implementation, including limited infrastructure, financial constraints, and a lack of digital literacy. This study aims to explore how local entrepreneurs can bridge these gaps and foster sustainable development. A mixed-methods approach was employed, combining qualitative interviews with 30 entrepreneurs and quantitative surveys distributed to 200 residents in targeted growth points. Data was analysed to assess the current state of technology adoption and the perceived role of entrepreneurs in facilitating smart initiatives. Preliminary results indicate that entrepreneurs are pivotal in driving technological innovation and community engagement. Many have initiated projects that enhance local governance, improve service delivery, and promote economic resilience. However, challenges such as inadequate support from government policies and limited access to funding hinder broader implementation. The study concludes that empowering entrepreneurs through targeted training, access to capital, and supportive policies is essential for the successful development of smart cities and villages in Zimbabwe. By leveraging technology, these entrepreneurs can significantly contribute to societal transformation, thereby aligning with broader national development goals. Future research should examine the long-term effects of these initiatives on community well-being and economic growth.

Key Words: [Smart Cities](#), [Smart Villages](#), [Entrepreneurs](#), [Technology](#), [Zimbabwe](#)



FRAMEWORK FOR CRYPTOCURRENCY ADOPTION AS A DIGITAL CURRENCY BY ZIMBABWEAN COMPANIES

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Zimbabwe has been experiencing mounting economic challenges for a long time such as high inflation, distrust in the banking industry, a confusing mix of different currencies, liquidity crunch and high transaction charges. Thus, its companies have been facing a tough business environment. Adoption of cryptocurrencies as a digital currency is a possible solution to most of the mentioned problems. Consequently, this study aimed to come up with a pragmatic cryptocurrency adoption framework to enable Zimbabwean companies and others to successfully adopt cryptocurrencies, such as Bitcoin, as a digital currency. The study population consisted of finance and senior managers of Zimbabwean companies trading in and not yet trading in cryptocurrencies. The participants considered cryptocurrencies to be a form of money. The major limitations to cryptocurrency adoption by Zimbabwean companies include a lack of a sound crypto-adoption framework, acute lack of knowledge about how cryptos and the blockchain technology work and their negative traits and high volatility. The study concluded that cryptos can be used as a digital currency. It made it clear how cryptos and the blockchain technology work. It explained how monetary authorities can promote the positive traits of cryptos while suppressing their negative traits. The common cryptocurrency adoption drawbacks have been exposed. The proposed framework allows governments to retain, to some extent, the monetary system control by instituting new forms of intermediaries. Adopting cryptos would likely go a long way in addressing the severe liquidity problems Zimbabwe has been facing for a long time. Wide crypto adoption by Zimbabwean companies will eventually enable Zimbabwe to blend the cryptos with physical fiat money. Early adoption of the cryptos will make Zimbabwean companies globally competitive and enables them to cheaply perform international financial transactions. The proposed framework informs individuals, companies and policy-making by the Reserve Bank of Zimbabwe (RBZ).

Keywords: of cryptocurrency, digital currency, crypto-adoption framework, blockchain technology, Zimbabwe



TRANSFORMING AGROTOURISM IN AFRICA: HARNESSING DIGITAL AND SOCIAL MEDIA MARKETING STRATEGIES FOR OVERCOMING BUSINESS CHALLENGES AMONG ZIMBABWEAN ENTREPRENEURS

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Africa's agrotourism scene is growing and has emerged as a promising sector to transform local economies by combining farming with tourism thereby offering unique opportunities for Zimbabwean entrepreneurs. Through leveraging agriculture and tourism there is a latent potential for economic development and transformation, especially in rural areas. However, in Zimbabwe, agrotourism businesses often struggle with problems like poor infrastructure, limited access to markets, poor visibility and difficulties in standing out from the competition. Digital and social media marketing have revolutionised and transformed the way businesses engage, promote and ensure visibility of their products or services. This research looks into how digital and social media marketing can help to can captivate potential visitors and generate interest in agritourism experiences as well overcome unique challenges faced by agritourism Entrepreneurs in Zimbabwe. It further explores common obstacles that agrotourism business face and discuss how using Digital and Social Media Marketing can help businesses build credibility, trust, reach more customers and grow. This can be achieved through using search engine optimization, creating relatable online content, and understanding customer behaviour through data analytics, all aimed at appealing to more people around the world. In addition, the research shares innovative ways digital technology, like mobile phones and online services, can make it easier for these businesses to connect with customers and handle sales. By including stories from successful Zimbabwean entrepreneurs, this research reveals how effective digital marketing can be in making agrotourism more popular and visible to tourists. Looking towards the future, we outline steps for maintaining a strong business and staying ahead of the game. This includes being smart with data, engaging with the community online, and improving digital skills. Overall, the research serves as a guide for using digital tools to refresh and sustain Africa's agrotourism, with a focus on Zimbabwe's experiences as an example for others.

Keywords: [Agrotourism](#), [Digital Marketing](#), [Social Media](#), [Entrepreneurial Challenges](#), [Zimbabwe](#)



CHALLENGES-FACED-IN-IMPLEMENTATION-OF-GREEN-PROCUREMENT

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The research is more centred on procurement and the importance that has been placed upon it in the past decade. It aims at identifying eco-friendly means of procurement and focusing much on green procurement. The sole purpose of this study being to seek the challenges faced in the implementation of green procurement as this helps in organisational performance in tertiary institutions in Zimbabwe. The study goes further to explore facts and opinions that have been expressed by other predecessor researchers with regards to procurement and green procurement in the background to the study section. A statement to the problem is then formulated from which research objective and question come from. Contributions of other researchers on matters relative to the research being carried out are then cited in literature review. Methodology then gives in detail all the type of research, the procedure carried out in carrying out the research, in data collection and presentation. The researcher used quantitative methodology and descriptive analysis. Then the researcher analysed the data to get to the findings. The main findings showed the numerous challenges being faced in the implementation of green procurement. The findings of the research are important as they can help organisations identify and rectify challenges when manoeuvring the implementation of Green procurement in their organisations for sustainable economic development.

Key words: [green procurement](#), [sustainability](#), [economic development](#)



LEVERAGING DIGITAL MARKETING FOR INCLUSIVE GROWTH: SUSTAINABLE STRATEGIES TO BRIDGE THE DIGITAL DIVIDE IN ZIMBABWE.

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This paper explores the use of digital marketing to foster inclusive growth by aligning sustainable strategies that bridge the digital divide in Zimbabwe. Previous studies highlight how entrepreneurial activity in business has been affected because of the digital divide. In today's fast - evolving digital landscape, technology has become a powerful driver of economic growth, social development, and environmental sustainability. However, the benefits of digital transformation are not evenly distributed. Many marginalized communities, particularly in developing regions, face barriers to accessing and utilizing digital technologies. To achieve true sustainable development, it is crucial to ensure that digital transformation is inclusive, enabling all individuals and societies to participate in and benefit from the digital economy. With the advent of digital marketing (social media marketing and mobile marketing strategies), efficiency and competitiveness in businesses and small to medium enterprises can be enhanced. Future generations heavily rely on our decisions and policies implemented in the now. Bridging the digital divide is imperative to foster equitable digital transformation that benefits industry and commerce. The primary objective of this paper focuses on how digital marketing (social media marketing and mobile marketing) can be utilized to ensure the future of marginalized businesses and communities. The sub objectives are to: (1) Analyze the current state of digital marketing adoption in rural small to medium enterprises; (2) Identify barriers and opportunities in bridging the digital divide for rural small to medium enterprises; (3) Explore the role of digital marketing in expanding market access rural small to medium enterprises; (4) intersection of policy and private sector strategies to bridge the digital divide. The focus of this paper intersects digital marketing, sustainable development and inclusive digital transformation. Centering on digital marketing explores how targeted strategies empower rural small to medium enterprises; resultantly stimulating economic growth and proliferating inclusive participation in the digital economy. This paper will follow a concurrent mixed methods approach. Data will be collected combining semi- structured interviews and focus group discussions (FDGs) with rural small to medium enterprises. The study will adopt a pragmatic research philosophy, which allows for the

integration of both qualitative and quantitative research methods to address the research problem comprehensively (Creswell & Creswell, 2018). This approach is appropriate because quantitative data provides measurable insights into digital marketing adoption rates, internet penetration, and economic impact while qualitative data explores perceptions, challenges, and opportunities from stakeholders, offering deeper contextual understanding (Bryman, 2016). The sample includes rural SMEs using digital marketing and rural consumers from different regions of Zimbabwe. This paper is expected to provide empirical evidence of how digital marketing promotes inclusive growth to bridge the digital divide and for sustainable development in Zimbabwe's marginalized business communities. The findings are expected to generate information to equip policy makers, marketing practitioners and those in industry and commerce to come up with practical solutions that ensure inclusivity and ensure future generations are not affected. The results not only add to academic literature but proffer practical solutions and recommendations to create and maintain an inclusive digital economy.

Keywords: Digital marketing, Inclusive growth, Digital divide, Sustainable development, Rural SMEs, Zimbabwe



IMPACT OF BANK-FINTECH COLLABORATION ON FINANCIAL INCLUSION IN SUB SAHARAN AFRICA.

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The convergence of digital technologies with the dawn of the fourth Industrial revolution has created a fertile ground for innovation in financial services. This has led to the emergence of Fintech (Anifa et al,2022). Global adoption is rapidly increasing due to its disruptive nature and is largely embraced by participants who are underserved by traditional financial service providers (IMF,2024). According to the Global Fintech Index (2019), global investments in fintech are growing rapidly year by year owing to increased interconnectivity with digital revolution. The total value of investment skyrocketed from \$60,2 billion in 2017 to \$150,3 billion, representing a staggering 250% increase within just two years (Fintech Index,2019). This trend continued in 2021, with a report by the Economist Group (2021) indicating that one in five dollars invested by Venture Capital firms went towards fintech ventures during the first half of the year. The COVID-19 pandemic has also

accelerated digitalization within the financial sector, reducing the frequency of physical interactions and prompting customers to increasingly manage their financial needs online (Ernest and Young, 2021). This shift has led to evolving customer expectations. The World Bank (2022) noted that the wide spread adoption of smartphones significantly fueled the growth of fintech, a trend that intensified during the pandemic. This shift was accompanied by a notable increase in digital payments and efforts to enhance financial inclusion (World Bank, 2022). Ironically, a staggering 1.4 billion adults globally remain unbanked and another 2.8 billion are underbanked (World Bank, 2022). The majority of these people are the hardest to reach and more commonly women, poorer, less educated, and living in rural areas. Interestingly, a report by Boston Consulting Group and QED investors (2023) highlighted that while nearly 89% of adults own mobile phones almost half still rely on cash for major transactions. According to the World Bank's (2024) report, in Sub-Saharan Africa, the percentage of adults with bank accounts has more than doubled since 2011 reaching 49%. Over the five years from 2017 to 2022, nine out of 36 surveyed economies experienced significant growth in account ownership, with mobile money adoption being a key driver in most cases. South Africa witnessed approximately 15% points growth in this period and the majority of SSA countries recording less than 5%. Thus making Sub Saharan Africa a compelling case for studying how fintech can address financial inclusion challenges. The main objective of this study is to analyse how bank-fintech partnerships in Sub-Saharan Africa contribute to financial inclusion. The study also has secondary objectives of identifying the key characteristics and motivations driving bank-fintech partnership in SSA and to investigate the impact of bank-fintech partnership on overall accessibility of financial services for underbanked and underserved populations. The study will make use of econometric analysis as a methodology. The study is expected to proffer solution to policy makers to come up with policies that increase access to financial services, increase transparency and sustained, inclusive and sustainable economic growth.

Keywords: Fintech, Financial inclusion, Bank-fintech partnerships, Digital transformation, Sub-Saharan Africa



ASSESSING THE ATTRACTIVENESS OF AFRICAN COUNTRIES TO CHINESE FOREIGN DIRECT INVESTMENT USING NON-PARAMETRIC APPROACHES

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China has become one of the leading sources of FDI in Africa. Yet the distribution of Chinese FDI among the various countries in Africa is concentrated. This study aims to assess African countries' attractiveness to Chinese foreign direct investment using a non-parametric, linear programming model called Data Envelope Analysis. This approach can also determine the existing capacity to enhance FDI inflow within the current economic context. The analysis will be based on averaged annual data from 2005 to 2022 on Chinese foreign direct investment and the macroeconomic environment in 42 African host countries. Data Envelopment Analysis will be used to generate efficiency scores, which represent the attractiveness of a host country, relative to peers, to Chinese FDI. The efficiency scores will be used to generate a country ranking, based on which the best-performing countries will be identified for purposes of peer learning. The data envelopment analysis will also determine the amount of slack in each country's performance. Based on the slack analysis, we will determine which countries have room to optimize their attractiveness to Chinese FDI through fine-tuning of the policy mix as well as adjustments in their macroeconomic environment. This article falls within the scope of the first topic of the conference, which focuses on 'Regional, Transnational and Global Trade and Investment for Industrialization of Africa'. Our contribution to the literature on FDI is twofold: most studies on Chinese FDI focus on the determinants, but our approach analyses which countries attract more Chinese FDI given their economic context. Secondly, unlike previous studies, this study is able to determine the available potential to optimize FDI inflow with the prevailing economic environment.

Keywords: Chinese FDI, Africa, Data Envelopment Analysis, Investment attractiveness, Macroeconomic environment



STRENGTHENING CLIMATE RESILIENCE THROUGH DIGITAL WEATHER FORECASTING: A CASE STUDY OF TOMATO FARMERS IN SOUTH AFRICA

Sifiso Mdluli , Prof MJ Ngoepe-Ntsoane, Dr Andy Okw

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Climate variability poses significant risks to agricultural productivity, particularly for smallholder farmers in South Africa who often lack timely and accurate weather information to guide decisionmaking. This study examines the role of digital weather forecasting in strengthening climate resilience among tomato farmers, a crop highly sensitive to temperature, rainfall variability, and extreme weather events. The study followed a systematic review Preferred Reporting Items for Systematic Reviews (PRISMA) and meta-analyses guidelines as a methodology. The peer-reviewed literature from 2010-2025 was undertaken across multiple databases including ScienceDirect, Scopus, and Google Scholar where twenty-eight studies were included after applying inclusion criteria. The focus was on digital weather forecasting, climate resilience, and agricultural adaptation in South African contexts, with specific emphasis on tomato production systems. The study explores how farmers access, interpret, and apply digital weather forecasts delivered via mobile applications, SMS alerts, and online platforms to inform planting schedules, irrigation planning, pest and disease management, and post-harvest operations. Findings reveal that access to digital forecasts improves farmers' capacity to anticipate and respond to weather-related risks, reducing crop losses and enhancing productivity. However, the effectiveness of these tools is influenced by digital literacy, the reliability of forecast data, and infrastructural limitations such as mobile network coverage. The research also highlights the importance of integrating indigenous knowledge with scientific forecasting to improve trust and adoption rates. Recommendations include targeted training for farmers, strengthening partnerships between meteorological services and agricultural extension officers, and investing in user-friendly, localised forecasting systems. By showcasing the transformative potential of digital weather forecasting, this case study contributes to policy and practice aimed at building adaptive capacity, enhancing climate resilience, and ensuring sustainable livelihoods for tomato farmers in South Africa.

Keywords: Climate variability, Digital weather forecasting, Climate resilience, Tomato production, South Africa



THE FUTURE OF WORK: BALANCING AUTOMATION, REMOTE WORK, AND HUMAN-CENTRED SKILLS IN THE DIGITAL AGE

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The future of the world of work is going through tremendous changes as a result of growing focus on human centric skills, remote work and automation. This study looks at the complex implications to world economies, workforce and individual organisation that come out as a result of these shifts. Concerns about redundancy, re-skilling and job displacement have become the major issues in the world of work due to the emergence of automation and artificial intelligence. Concurrently the embracing of emerging concepts such as hybrid and remote working has revolutionised the workplace. This has raised questions of organisational culture maintenance, the wellbeing of employees as well as issues of productivity. This research uncovers how these emerging trends intersect making special reference to role of human centred skills such as creativeness, emotional intelligence and adaptability in manoeuvring through the current technology driven workplace. The research combines a mixed method approach making the use of quantitative analysis of trends in the job market as well as qualitative views from employees as well as leaders in the industry; the study uncovers ways of promoting inclusivity and resilience. In addition, the study analysis the sustainability and ethical issues brought by these changes and ultimately will lead to recommendations to educational institutions, business as well as policy makers. Through the addressing of the intersection of remote work, automation and human skills this research aims to create an adaptive, sustainable and human centred work environment that caters for the rapidly changing business and work environment.

Key words: Future of work, Automation, Remote Work, Human Centred skills and Digital revolution



BUSINESS INTELLIGENCE FRAMEWORK ON ELECTRONIC BANKING IN THE ZIMBABWE BANKING SECTOR.

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Despite the growth of electronic banking in Zimbabwe's financial services sector, empirical evidence from literature indicates lack of coherence on business intelligence and electronic banking integration. The absence of a framework on the nexus of business intelligence and electronic banking compounds the challenge. The researcher adopted Pragmatic Paradigm which focuses on what works rather than what was adopted and objectively true or real. Pragmatic Paradigm advocates Mixed Methods Approach which uses both the quantitative and qualitative data collection methods. In addition, the study adopted the Sequential Exploratory Research Design. The Research targeted all commercial banks in Zimbabwe which have total population of 4,000 employees according to the RBZ Economic Bullen issued in March 2023. The researcher sampled 381 participants who completed questionnaires and 9 interviewed. The study regressed the quantitative data from questionnaires establishing the relationship among electronic banking variables and business analytics variables. Furthermore, the researcher thematically analysed qualitative data from interviews. The study integrated the results showing that business intelligence variables correlated with electronic banking variables. Hence, the study recommends the adoption of Business Intelligence Framework that promote electronic banking for the sustainability of Zimbabwe financial service sector.

Keywords: [Business Analytics](#); [Financial Services](#), [Business Intelligence](#), [Electronic Banking](#)



SKILLS, EDUCATION, EMPLOYMENT, AND SUSTAINABILITY IN RENEWABLE ENERGY SMES: A NAMIBIA-ZIMBABWE CROSS-COUNTRY COMPARISON

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This study aims to (1) Assess the alignment between national education systems and the skills demanded by renewable energy SMEs in Namibia and Zimbabwe (2) Evaluate the extent to which current vocational and STEM curricula address critical competencies focusing on digital literacy, technical proficiency and problem-solving. (3) Investigate the impact of skills gaps on SME productivity, survival rates, and regional disparities (4) Analyze urban-rural divides in access to finance, training, and infrastructure for renewable energy SMEs. A mixed-methods approach is employed, combining primary data, secondary data and regional comparisons. This is achieved by firstly, conducting surveys of 400 SME owners (200 per country) and 600 employees (300 per country) conducted in 2025. Secondly, performing national labor force surveys, World Bank reports, and policy documents. Thirdly, conducting focused case studies between Zimbabwe's Masvingo province and Namibia's Omaheke region to contextualize rural SME challenges. Quantitative data is analyzed for trends in skills mismatches and productivity, while qualitative insights explored systemic barriers and localized growth constraints. This study contributes to policy by coming up with proposals for vocational education reforms, block chain-based credentialing systems, and SADC-supported financing to align academia with industry needs. Furthermore, the study comes up with a regional collaboration framework which incorporates strategies to bridge urban-rural disparities and foster cross-border partnerships between academia, SMEs, and policymakers. The study has practical relevance in that it provides evidence-based solutions to enhance SME resilience, directly supporting the conference's goal of inclusive growth through academia-industry-commerce synergy. Additionally, the study contributes to sustainable development alignment by providing a roadmap for renewable energy SMEs to drive equitable economic transformation, emphasizing digital inclusion and sustainable practices.

Keywords: SMEs, skills mismatch, vocational training, regional disparities, Namibia-Zimbabwe.



AI-ENABLED CROP PLANNING FOR SMALLHOLDER FARMERS: ENHANCING PRODUCTIVITY AND CLIMATE ADAPTATION IN SOUTH AFRICA

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The incorporation of artificial intelligence in agriculture has the potential to mitigate against various challenges caused by climate change by providing data-driven insights and improving decision making in crop planning. Smallholder farmers in South Africa face many challenges in the adoption of artificial intelligence due to lack of resources, digital illiteracy and lack of access to market information. This study investigates South African smallholder farmers adoption of artificial intelligence in crop planning to mitigate against the effects of climate change and improve productivity. The study's objectives were to investigate how AI-enabled crop planning enhanced productivity and climate adaptation among smallholder farmers in South Africa; the opportunities that for AI tools presented in enhancing resource management, crop selection and planting schedules as well as the challenges faced by smallholder farmers in adopting AI-enabled crop planning. The methodology involved a systematic review of existing academic literature derived from Google Scholar, Scopus and Web of Science with inclusion and exclusion criteria used to select the most relevant articles. The findings provide practical recommendations for agricultural practitioners and policymakers and to enable to adoption of AI tools for smallholder farmers in South Africa.

Keywords: Artificial intelligence, Crop planning, Climate change adaptation, Smallholder farmers, South Africa



THE EFFICACY OF TECHNICAL VOCATIONAL EDUCATION AND TRAINING PROGRAMS IN ALIGNING GRADUATE SKILLS WITH THE DEMANDS OF AFRICA'S EMERGING GREEN ECONOMY: A LITERATURE REVIEW.

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This literature review aims to compile available studies on the present state of TVET programs in equipping graduates for the fast-growing green economy throughout the African continent, hence tackling the central theme of "Future of Work: Skills, Education, Employment, and Small Business Development in Africa," and particularly addressing the research paper entitled "The Efficacy of Technical Vocational Education and Training (TVET) Programs in Aligning Graduate Skills with the Demands of Africa & its Emerging Green Economy." The goal of this study is to address existing research skill gaps, assess how well the TVET program is working, and propose strategies for improving alignment between training and industry requirements. Using a systematic literature review approach, this research examines many academic papers, economic studies, and policy documents from African governments and international organisations. Emphasizing their capacity to provide skills pertinent to industries including renewable energy, sustainable agriculture, and eco-tourism, the review rather closely analyzes the available content of current TVET programs. Significant key results emphasize a significant misalignment in advanced technical abilities, entrepreneurial flair, and ability to adapt to changing technology between present TVET results and the specific skills required by the green economy, particularly in industries. The review emphasizes the need for industry partnerships, curriculum updates, and teacher training by highlighting effective program models and best practices in Africa. By offering a thorough examination of the problems and possibilities in green skills development across Africa, this study adds to the conference. It provides great ideas for government officials, teachers, and industry players who want to improve TVET programs and support a skilled labor force for the green economy. Recommendations cover the establishment of strong industry-academia partnerships; the inclusion of digital literacy and green technology modules; and the integration of hands-on, practical training. This review also calls for the creation of regional and national policies that give green skills development top priority and encourage sustainable economic development. Future studies using empirical data should evaluate

the effect of TVET initiatives on graduate employability and the expansion of the green economy in many African settings over the long term.

Keywords: Africa; Technical Vocational Education Training; Curriculum; Green economy; Industry partnerships.



BRIDGING THE DIGITAL DIVIDE: AN ASSESSMENT OF E-LEARNING ACCESSIBILITY AND IMPACT ON SKILL DEVELOPMENT FOR RURAL YOUTH IN ZIMBABWE'S SECONDARY EDUCATION SECTOR

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This research addresses the conference theme, "Future of Work: The study explores the conference theme "Future of Work: Skills, Education, Employment, and Small Business Development in Africa" through an analysis of the digital divide affecting rural secondary education in Zimbabwe. Specifically, it focuses on the topic entitled: "Bridging the Digital Divide: An Assessment of E-Learning Accessibility and Impact on Skill Development for Rural Youth in Zimbabwe's Secondary Education Sector." The study evaluates digital learning access and its effects on skill advancement for young students across Zimbabwe's rural primary and secondary educational institutions. This study aims to review the accessibility of e-learning platforms while identifying implementation barriers and measuring how current e-learning projects affect rural youth's skill development. This research uses a mixed-methods approach that incorporates both quantitative and qualitative data collection techniques. The quantitative data from teachers, and parent surveys in selected rural schools evaluate access to digital devices and internet connectivity along with digital literacy levels. Semi-structured interviews and focus group discussions with purposively selected educational stakeholders generated qualitative data which offer detailed perspectives on both the challenges and opportunities presented by e-learning implementation. The study analyzes how student performance relates to e-learning access and examines how students develop necessary digital competencies. The study presents empirical findings at the conference about how the digital divide affects educational outcomes within an African setting. The study showcases unequal access to e-learning resources and emphasizes the necessity of specific strategies to provide fair digital education options for young people living in rural areas. E-learning effectiveness faces major

obstacles from poor infrastructure, insufficient teacher training and limited financial resources. Investment in internet infrastructure alongside affordable digital devices and context-specific digital literacy programs paired with teacher training in digital pedagogy are recommended for improvement. The study recommends the development of local digital centres along with public-private partnerships as strategies to close the digital gap and equip rural youth for future employment opportunities to expand their career options.

Keywords: Digital divide; E-learning; Skills Development; Primary and secondary education; Digital pedagogy.



INNOVATIVE AGRO-PROCESSING TECHNOLOGIES AS CATALYSTS FOR RURAL INDUSTRIALIZATION AND ECONOMIC GROWTH. "A CASE STUDY FOR MARANGE IN MUTARE.

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This study investigates the role of context-specific agro-processing technologies in catalyzing rural industrialization and enhancing agricultural value chains in Marange, a semi-arid region in Zimbabwe's Mutare District. Facing persistent challenges such as high post-harvest losses, limited mechanization, and feed shortages, smallholder farmers often rely on labour-intensive and inefficient traditional methods. In response, three low-cost, manually operated innovations were designed and field-tested: a hybrid small grain thresher-winnower, a hand-operated poultry feed mill, and a silage-making machine. These technologies were developed through a participatory, TVET-integrated approach that merged technical education with grassroots innovation, utilizing design tools like Quality Function Deployment (QFD), TRIZ, SCAMPER, and black-box modelling. The machines were fabricated at Kwekwe Polytechnic and tested with over 100 farmers during field trials between May and July 2024. Results demonstrated substantial improvements: the thresher achieved 94% efficiency with a 69.2 kg/hr output, the feed mill delivered 48 kg/hr with minimal energy input, and the silage maker processed 300 kg/hr with 93% compaction efficiency. The technologies reduced labour burdens, increased grain and feed quality, and enabled off-grid operation benefiting rural farmers. Additionally, the initiative fostered hands-on engineering education and aligned with rural entrepreneurship goals. This study offers a replicable model of education-industry collaboration and

gender-sensitive design to advance rural mechanization, food security, and inclusive economic development in drought-prone regions.

Keywords: Agro-processing, Rural Industrialization, TVET, Smallholder Farmers, Feed Mill, Silage, Small Grain Threshing, Innovation



FINANCIAL INCLUSION OF RURAL YOUNG WOMEN IN ZIMBABWE: A CASE OF SAVING CLUBS IN WARD 2 IN CHIREDDI.

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Most of the rural poor women around the world remain excluded from the financial system, limiting their access to savings, credit, and insurance services. The current study sought to explore the financial inclusion of rural young women in ward 2 in Chiredzi. Young rural women in Chiredzi tend to disproportionately lack access to financial services and credit which is a major contributing factor that perpetuates poverty among them. The study followed a qualitative research approach and a phenomenological design. The target population for the study was rural women aged between 18 and 35 years, who belong to a savings club in ward 2 in Chiredzi. Purposive and snowball sampling technique were used to select study participants and a sample of ten participants made up of young rural women, officers from Ministry of Women Affairs and Economic Development Officer in the District Development Coordinator's Office. Key informant interviews were conducted to gather the stakeholders' views on the subject matter. Thematic data analysis was used in the study. The findings showed that women fail to join savings clubs due to inability to pay membership fees. Poverty, lack of capital and inability to sustain savings and lending scheme's expectations discourages most of those that are poor and marginalized. Through collective action, savings clubs contribute to socioeconomic change, poverty reduction, and inclusive development. Women's economic empowerment increases women's access to economic resources and opportunities including jobs, financial services, property and other productive assets, skills development and market information. The study concluded that saving groups have given women the power to take control over their everyday life in a way that they previously not possible. Access to money is found to empower women

by enabling them to independently meet their social needs, enhance decision making and promote inclusive development. The study recommended that for the purpose of sustainability of livelihood of the rural women, village savings and loan groups must train their members on benefits of saving and taking loans and this could enhance the growth of this trend. Social workers can analyse the situation of women in rural areas with a view to identifying priority areas where positive action is needed.

Key words: Financial inclusion; rural young women; savings clubs



THE ROLE OF INTELLECTUAL PROPERTY IN DRIVING THE OPERATIONALIZATION OF EDUCATION 5.0 IN TVET INSTITUTIONS: THE CASE OF KWEKWE POLYTECHNIC, ZIMBABWE

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This study critically examined the role of Intellectual Property (IP) in advancing the implementation of Zimbabwe's Education 5.0 framework at Kwekwe Polytechnic, with a focus on fostering innovation, industrialization, and technopreneurship as espoused in the Education 5.0 doctrine. To drive economic development, Education 5.0 integrates teaching, research, community engagement, innovation, and industrialization in TVET aligned higher education training centers. However, the absence of strong intellectual property (IP) frameworks and IP ecosystems in Technical and Vocational Education and Training (TVET) institutions impedes intellectual output protection and commercialization of research and development outputs in the form of goods and services. This study objectives were as followings: to assess the awareness and understanding of intellectual property (IP) among students, faculty, and administrators at Kwekwe Polytechnic. To explore the influence of IP policies on promoting innovation, industrialization, and entrepreneurship within the framework of Education 5.0. to identify gaps in the existing IP frameworks at Kwekwe Polytechnic that may hinder the operationalization of Education 5.0 and to develop actionable recommendations for enhancing IP awareness, policy implementation, and institutional practices to support Education 5.0. The research method used considered the concurrent triangulation research design and enabled collection of data from 80 respondents that included students, lecturers and administrators. A valid response rate of 67, 5% was obtained. The findings show an inconsiderable or inadequate awareness

of intellectual property rights such as copyrights and trademarks, with significant knowledge gaps regarding patents, utility models, industrial designs and trade secrets. The study revealed the need for comprehensive and enforceable intellectual property frameworks to enable alignment of TVET training programmes with national economic imperatives. The study suggests strengthening intellectual property policy frameworks, investing in targeted capacity-building programs, and fostering collaborative industry partnerships to improve TVET institutions' innovation ecosystems. Future comparative research is proposed to generalize findings and identify best practices across multiple institutions, thereby advancing Zimbabwe's sustainable innovation-driven education system.

Keywords: Intellectual Property, Technopreneurship, Utility models, Patents, IP Policy, Education 5.0, Innovation.



FOSTERING MANUFACTURING ENTREPRENEURSHIP THROUGH POLYTECHNIC- INDUSTRY COLLABORATIONS: A PATHWAY TO ECONOMIC GROWTH IN ZIMBABWE

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Manufacturing entrepreneurship is vital for industrialization, economic growth, and job creation in Zimbabwe. However, the sector faces challenges such as limited technical expertise, inadequate funding, weak policies, and poor industry-academic collaboration. Polytechnic institutions, with their hands-on technical training, could help bridge these gaps by fostering innovation, skills development, import substitution and entrepreneurship. However, weak industry-polytechnic partnerships have hindered knowledge transfer, research commercialization, and business growth. Strengthening these collaborations could enhance manufacturing entrepreneurship, increase industrial productivity, and drive economic development. To align technical education with industrial needs, the Zimbabwean government, in 2005, mandated polytechnics to specialize in specific industry-relevant fields. Harare Polytechnic was assigned Chemical Engineering, Mutare Polytechnic Wood Technology, Bulawayo Polytechnic Civil Engineering, Kwekwe Polytechnic Heavy Engineering, and Masvingo Polytechnic Food Processing. While this initiative aimed to align polytechnic education with national industrial

needs and enhance localised industrialisation, its impact on manufacturing entrepreneurship and industrial development remains limited. This research was underpinned by Schumpeter's Theory of Innovation, which emphasized technological progress as a driver of economic development, industry, and government, Human Capital Theory, which examined how polytechnic education enhanced entrepreneurial skills. Additionally, University Theory of Economic Development was applied to assess how polytechnics contribute to economic growth through research, knowledge dissemination, and industry engagement. The study sought to achieve four primary objectives; to evaluate the contribution of polytechnic institutions to manufacturing entrepreneurship in Zimbabwe, to analyse existing industry-polytechnic collaborations and their impact on business development, to identify barriers hindering effective partnerships, and to propose strategies for strengthening industry-academic linkages to drive innovation, skills development, and economic growth. A mixed-methods approach was employed, incorporating both quantitative and qualitative techniques. Surveys were conducted among polytechnic students, lecturers, and industry representatives to assess their perspectives on existing collaborations. In-depth interviews with policymakers, entrepreneurs, and institutional leaders provided deeper insights into challenges and opportunities. Additionally, case studies from developed countries were analysed to extract lessons and best practices. Quantitative data was examined using statistical methods, while qualitative data underwent thematic analysis to identify key patterns and trends. Preliminary findings indicate that while Zimbabwe's polytechnics have significant potential to drive manufacturing entrepreneurship and promote import substitution, they face constraints such as outdated curricula, financial limitations, and weak industry engagement. Existing collaborations are often informal and lack structured mechanisms for technology transfer, joint research, and business incubation. However, successful models from countries like Germany, China, and South Korea suggest that well-organized industry-polytechnic partnerships can enhance innovation, industrial efficiency, and small business growth. This study aligns with the ATLAS Conference themes on academia-industry collaboration for economic transformation. Expected contributions include structured frameworks for stronger industry-polytechnic linkages, recommendations for policy and funding improvements, and enhanced strategies for research commercialization. Furthermore, the study will provide models for start-up incubation, technology transfer, and business mentorship within polytechnics.

Strengthening these partnerships will support industrial expansion, increase employment opportunities, and enhance economic diversification. Ultimately, this research aims to provide a roadmap for leveraging academia-industry collaboration to foster sustainable industrial growth and economic development in Zimbabwe.

KeyWords: Manufacturing entrepreneurship; Industry-academia partnership ; Localized industrialization



TECHNOLOGY'S IMPACT ON INNOVATION IN EDUCATION 5.0: NAVIGATING CHALLENGES OF INTELLECTUAL PROPERTY IN A DIGITAL AGE – CASE OF KWEKWE POLYTECHNIC, ZIMBABWE

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This research investigates the challenges faced by Kwekwe Polytechnic in navigating intellectual property (IP) issues within the context of Education 5.0 and rapid integration of technology. The study examines the impact of digital technologies on innovation in educational practices and the subsequent implications for IP protection. The study used the following objectives: to assess the current state of IP awareness among staff and students at Kwekwe Polytechnic, to identify the key challenges faced by the institution in managing IP in the digital age. The study evaluates the effectiveness of Kwekwe Polytechnic's existing IP policies and procedures. It then proposes recommendations for enhancing IP management and protection within the context of Education 5.0. The study adopted a mixed-methods approach and analyses the institution's existing IP policies, awareness levels among staff and students, and the practical challenges encountered in safeguarding digital innovations. The concurrent triangulation research design and abductive approach was used. The sample was selected using the stratified sampling, purposive and convenience sampling techniques. The total population targeted was 130 people and from that total, 97 people became the sample size from which data for this research study was collected. A combination of online questionnaires and interviews were the used research instruction. A valid response rate of 71.2 % was obtained. Findings reveal a gap between the institution's IP framework and the dynamic nature

of digital innovation, highlighting the need for enhanced IP awareness, robust policy implementation, and strategic resource allocation, challenges in protecting digital educational resources and a need for enhanced training and support. The study recommended several strategies to address the evolving challenges in intellectual property rights due to digital advancements. These included developing comprehensive IPR policies tailored to the digital education context, implementing IPR training and awareness programs for Administration, Divisions and students, establishing clear guidelines for the use and distribution of digital educational resources, promote a culture of ethical digital resource utilization and suggest updates to Zimbabwean IPR law to match the digital age. Area for further research should look at how the use of Open Educational Resources (OER) within Education 5.0 affects IP rights and responsibilities for both educators and students. **Contribution:** It advised policymakers to harmonize IP standards globally and support innovations in IP protection technologies. The study recommended legal reforms to update existing laws to better suit the digital age. The establishment of an IP management office, and the adoption of digital tools for IP asset management. It suggested the adoption of technologies like block chain, virtual reality and augmented reality for better IP management and the need for improved international cooperation to address cross-border enforcement issues.

Keywords: Education 5.0, Intellectual Property, Digital Age, Innovation, Technology, Block-chain, Virtual Reality, Augmented Reality



SUSTAINABLE DEVELOPMENT AND INCLUSIVE DIGITAL TRANSFORMATION

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In Zimbabwean context, the poultry industry plays a significant role in the country's economy, providing a vital source of income and food security. Most small scale poultry farmers are shifting from broiler rearing to road runner rearing due to production costs. Farmers are finding it cheaper to incubate fertilised eggs for chick production. Small scale poultry farmers prefer to store eggs so that the eggs accumulate in order to best utilise hatchery facilities. The major challenge is on pre-

incubation embryo development due to unfavourable parameters recorded in the rooms in which the fertilised eggs are stored prior to incubation. The concept of egg storage unit is common in large scale poultry plants in Zimbabwe. It is against this background that the idea of producing a hybrid powered egg storage unit which is not affected by external conditions. It is suitable for both urban and rural farmers due to it being electricity and solar powered. The aim was to design and construct a pre incubation egg storage unit in the form of a prototype which can house 100 eggs. This is now being develop to an upward sensitive scale as per poultry farmer's desired capacity. Best possible components as well as cubicle material were selected through a concept development approach. Parameters were calculated using appropriate formulae. By use of proteus 8 professional software, an electric circuit to monitor and control the parameters was deigned. Communication protocols were also developed to enable real time remote monitoring and conveyance of data collected from the pre- incubation unit. The refridgeration system demonstrated effective temperature and humidity regulation, oxygen and carbondioxide detection and automated egg turning functionality. The project showcases the integration of hardware and software engineering to create a robust and reliable solution for the small-scale poultry farmer. It is through digitalisation that this new solution to mitigate risk from unfavourable external conditions has been developed. Hence leveraging digitalisation to improve poultry production.

Keywords: Poultry industry, Egg storage unit, Hybrid power system, Small-scale farmers, Zimbabwe



CLIMATE CHANGE AND PEST MANAGEMENT: A SYSTEMATIC REVIEW OF DIGITAL TOOLS, INTEGRATED STRATEGIES, AND CLIMATE-DRIVEN PEST OUTBREAKS IN SOUTH AFRICA

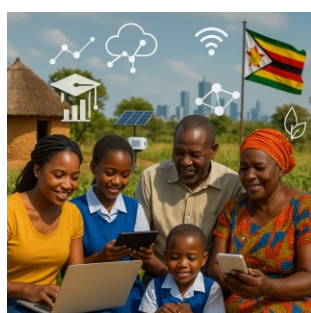
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Climate change is increasingly influencing agricultural pest dynamics, leading to more frequent, severe, and geographically widespread pest outbreaks that threaten global food security. This systematic review synthesizes recent literature (2018–2025) on how climate variables alter pest population growth, distribution, and outbreak patterns. It critically examines the emergence and application of digital pest detection technologies such as artificial intelligence, remote sensing, and mobile applications, assessing their potential to enhance early warning and precision pest

management. The review further explores how Integrated Pest Management (IPM) systems are adapting through climate-smart approaches, including the incorporation of ecological resilience, climate forecasting, and decision support tools. Despite technological and strategic advances, challenges such as pesticide resistance, reduced chemical efficacy, socio-economic constraints, and data gaps limit effective management. The review concludes that an integrated approach combining digital innovation, adaptive IPM, farmer capacity building, and supportive policy frameworks is essential to build resilience and sustain agricultural productivity under ongoing climate change.

Keywords: Climate change, Pest dynamics, Digital pest detection, Integrated Pest Management, Food security



INCLUSIVE DIGITAL TRANSFORMATION FOR SUSTAINABLE DEVELOPMENT TO BRIDGE THE DIGITAL DIVIDE IN SUB-SAHARAN AFRICA: A CASE STUDY IN ZIMBABWE

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As Sub-Saharan Africa (SSA) undergoes a rapid digital transformation, it is crucial to ensure that digital development is inclusive, supporting sustainable growth. Zimbabwe is a notable case study illustrating the potential benefits and challenges of aligning digital progress with the United Nations' Sustainable Development Goals (SDGs). Despite significant progress in mobile banking, e-governance, and digital entrepreneurship, the country still lags in addressing challenges such as inadequate infrastructure, digital literacy inequalities, and regulatory hurdles, which hinder equitable participation in the digital economy. This study examines the role of inclusive digital policies, new technologies, and public-private sector collaborations in promoting economic growth, social inclusion, and environmental sustainability. Through a study of Zimbabwe's EcoCash mobile money, the Smart Zimbabwe 2030 digital master plan, and efforts in e-agriculture, the research sheds light on the key drivers of digital inclusion while also acknowledging prevailing challenges, including high data prices, the urban-rural divide in digital technology, and the risk of cybersecurity threats. Additionally, the paper assesses the potential of AI-based education, blockchain technology, and green ICT innovations to enhance financial inclusion and promote digital equity, thereby empowering vulnerable groups. As Sub-Saharan Africa undergoes rapid transformation towards digitalisation, the region faces a paradox of progress paired with continued marginalisation. While

Zimbabwe has made notable strides in mobile banking, e-governance, and digital entrepreneurship, it still faces significant infrastructure deficits, digital literacy gaps, and regulatory challenges that hinder inclusive participation in the digital economy. Exorbitant data prices, inequalities in access to digital services between urban and rural areas, and cybersecurity threats exacerbate disparities, depriving marginalised groups of the benefits associated with the digital revolution. Without responsive policies, strong infrastructure, and efficient public-private partnerships, the path to digital development will likely widen existing socioeconomic gaps instead of mitigating them. To evaluate the effectiveness of inclusive digital policies, emerging technologies, and strategic partnerships in bridging Zimbabwe's digital divide and fostering sustainable socioeconomic development. This study employs a mixed-methods approach, combining qualitative and quantitative methods, to ensure a comprehensive investigation of the digital transformation landscape in Zimbabwe. Qualitative Analysis - Conducting case studies on Zimbabwe's digital economy initiatives, including EcoCash, Smart Zimbabwe 2030, and e-agriculture projects. Semi-structured interviews with policymakers, technology providers, and community stakeholders will provide insights into digital inclusion efforts and regulatory challenges. Quantitative Analysis - Conduct surveys and collect statistical information regarding levels of digital literacy, adoption rates of mobile money, and the spread of ICT infrastructure to assess the effectiveness and coverage of digital policy interventions. Comparative Review - A review of Zimbabwe's progress in digitalisation compared to other nations in Sub-Saharan Africa to determine best practices and replicable solutions. Initial findings suggest that for digital transformation in Zimbabwe to be inclusive and sustainable, it must adopt agile regulatory environments, invest in robust digital infrastructure, and expand digital literacy programs. Additionally, integrating renewable energy technologies into ICT infrastructure can enhance resilience to climate change and improve connectivity in areas without grid connections. The study concludes with policy recommendations emphasising the need for collaborative partnerships among stakeholders, new financing arrangements, and digital rights protection as key elements in bridging the digital divide and fostering sustainable, long-term development.

Keywords: Digital Inclusion, Sustainable Development, Sub-Saharan Africa, Zimbabwe, Digital Economy, E-Governance, Mobile Money, Green ICT, SDGs, Digital Divide.



THE CONTRIBUTION OF DIGITAL SKILLS AND KNOWLEDGE TO UNIVERSITY STUDENTS' INCLINATION TO EMBARK ON THE FASHION AND BEAUTY DIGITAL ENTREPRENEURIAL START-UPS IN SOUTH AFRICA: A SCOPING REVIEW

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Digital skills and knowledge of aspiring entrepreneurs have emerged as pivotal contributors to their inclination to start up digital enterprises. In the contemporary digital economy, the fashion and beauty industry has taken a lead in moving their entrepreneurial activities from legacy media platforms to the digital space especially by the youth as consumers and entrepreneurs. Despite digital transformation, fashion and beauty enterprises face considerable hurdles in leveraging digital skills for sustainable start-ups, particularly in the global market. This study explored the question: How do digital skills and knowledge contribute to university students' inclination to embark on the fashion and beauty digital entrepreneurial start-ups in South Africa? Methodologically, this study adopted an interpretivist lens utilising a scoping review of literature from selected scholarly databases. The study also utilized a qualitative research approach to explore students' digital skills and knowledge to embark on a fashion and beauty entrepreneurial start-ups and presents the search and selection process using a PRISMA flow diagram utilizing a Boolean search strategy. The inclusion criteria for the study encompassed publications with a specific year range (2020 – 2025), written in English, peer reviewed, and relevant to the focus on digital skills, entrepreneurship, university students within the specified context, while studies were excluded if they were not written in English, not peer reviewed, unrelated to fashion and beauty sector, and not focusing on the South African context. The study findings reveal that the inclination to embark on fashion and beauty digital entrepreneurial journey is influenced by the attainment of digital skills and knowledge. Among the core skills for digital entrepreneurship identified were knowledge of running a website, data analytics, digital marketing and cyber security. It was recognised that the entrepreneurial education contributed to attaining digital entrepreneurial skills and knowledge, however, it is not inadequate to sufficiently equip them for their entrepreneurial journeys in the dynamic and complex technological landscape. We recommend that university students exploit the numerous programmes available to equip them with requisite digital skills and knowledge.

Keywords: digital skills and knowledge, digital entrepreneurship, university students, fashion and beauty, digital entrepreneurial start-up, South Africa, entrepreneurial sustainability.



PUBLIC-PRIVATE PARTNERSHIP AND INDIGENOUS KNOWLEDGE PRODUCTION FOR AFRICAN DEVELOPMENT

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Partnership and collaboration have found themselves into academic research and knowledge production in recent time. This strategic innovation remains the onions of public-private partnership gearing towards collaborating and finding lasting solution to imminent challenges in the world. Having identified the strategic roles of public-private initiative in addressing developmental challenges, the drive for indigenous knowledge production for African development could also be located within this model to establish a contemporary synergy. This study will dive into the exploration of this model to underscore a reality that is currently blowing African nations on the need to fully embark on endogenous intervention that is more ecological to rescue the continent from her developmental doldrum. It is an attempt to promote a shift from the existing status quo by championing a novel ideal which embrace a sponsored knowledge production industry in our higher institutions for the purpose of continental transformation. This article will employ qualitative approach and rely heavily on critical and descriptive analytical frameworks potent for a value driven analysis. As part of the effort to deconstruct the over-reliance on exogenous approach to development in Africa, this article shall contribute to the conference's debate for a town-gown collaboration for African growth, bringing the PPP model to translate African knowledge economy from ordinary raw material continent to drive her development. Also, the study tends to decongest the rentier economic paradigm dominating the African economic analysis and cause a distinction that there are African indigenous capitalists whose efforts could be joined with the public sector interventions for the ultimate goal of African's social, economic, political, technological, and cultural transformation.

Keywords: PPP, indigenous knowledge, production, development, Africa



ENTREPRENEURSHIP AS A CATALYST FOR EMPLOYMENT: FOSTERING SMALL BUSINESS GROWTH IN SOUTH AFRICA.

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Entrepreneurship plays an increasingly pivotal role in stimulating economic development and catalyzing employment opportunities across South Africa. This paper explores how entrepreneurial initiatives foster small business growth, thereby addressing persistent challenges of unemployment and poverty in diverse South African contexts. Drawing upon case studies, and economic data, the study highlights the transformative impact of entrepreneurial ventures in mobilizing local resources, driving innovation, and promoting sustainable employment. The research examines the interplay between supportive policy frameworks, access to finance, and entrepreneurial training programmes that collectively empower individuals to launch and scale small businesses. Findings indicate that entrepreneurial activity not only generates jobs but also cultivates dynamic business ecosystems that spur community development and enhance competitiveness in local and global markets. Moreover, the study reveals that entrepreneurship contributes to gender inclusivity and youth empowerment by providing alternative avenues for economic participation and leadership. Despite challenges such as limited access to capital, infrastructural deficits, and regulatory constraints, entrepreneurial efforts have demonstrated significant resilience and adaptability in meeting evolving market demands. This analysis underscores the importance of targeted interventions, including policy reforms and capacity-building initiatives, to nurture a conducive environment for entrepreneurship. By fostering small business growth, entrepreneurship emerges as a critical catalyst for employment and economic diversification in South Africa, paving the way for long-term socio-economic transformation and sustainable development. This comprehensive examination not only elucidates the multifaceted role of entrepreneurship in driving job creation and economic resilience but also offers insights for policymakers and investors seeking to harness Africa's potential for sustainable growth.

KEYWORDS: Entrepreneurship, Employment Generation, Small Business Growth, Economic Development, South Africa.



EXPLAINABLE ARTIFICIAL INTELLIGENCE IN HEALTHCARE: APPLICABILITY IN AFRICAN COMMUNITIES

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Explainable AI (XAI) has the potential to revolutionize healthcare by improving the transparency, dependability, and ethical compliance of AI-driven medical decisions. Despite its potential, the healthcare industry still faces a number of obstacles, such as integrating XAI into clinical workflows, maintaining strict regulatory compliance, and striking a balance between interpretability and accuracy. These challenges are more common in the African healthcare facilities mainly due to infrastructural challenges.

This paper is based on secondary data and systematic review on emerging technologies in Africa is done. Different AI initiatives across Africa are considered. The objective of the paper is to assess Explainable Artificial Intelligence in Healthcare and its applicability to African continent. The understanding and practical application of XAI in healthcare are improved by this paper's thorough review of the field, which covers methods, difficulties, opportunities, and developments. The study also examines ethical AI in healthcare, addressing fresh viewpoints and developing patterns while providing insightful information for practitioners and researchers. By guiding future research and policy-making, the insights and suggestions offered hope to promote the creation of open, reliable, and efficient AI-driven solutions. Available results have shown that XAI enhances telemedicine and remote diagnostics' accuracy and accessibility, especially in underprivileged areas. XAI helps remote healthcare providers comprehend and trust AI recommendations by making sure that AI-driven diagnostic suggestions are accompanied by clear explanations, improving care quality in situations where access to specialists is limited. In addition to being essential for increasing diagnostic precision, XAI in telemedicine is also essential for fostering trust between patients, healthcare professionals, and AI tools in remote locations. The paper informs and recommends the best approaches for African communities to benefit for emerging technologies within the healthcare. The paper provides insights on AI applications within the healthcare sector within the African continent. The paper fits well under the technology enabled healthcare subtheme of the conference. Additionally, some elements of digital inclusion, digital transformation and societal impact of technologies are covered within the paper.

Keywords: Explainable AI (XAI), AI, bias, ethics, fairness, healthcare, machine learning,



OVERCOMING OBSTACLES AND MAXIMIZING OPPORTUNITIES FOR AI-DRIVEN RESEARCH IN ZIMBABWE'S INDIGENOUS LANGUAGES: PATHWAYS TO INNOVATION AND INCLUSIVE SUSTAINABLE ECONOMIC GROWTH

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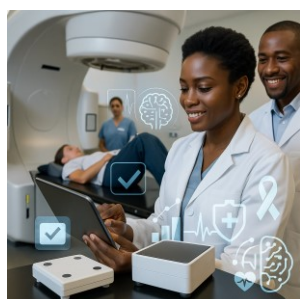
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The integration of Artificial Intelligence (AI) into research in Zimbabwe's Indigenous languages presents significant challenges, immense potential and transformative opportunities for fostering innovation and inclusive sustainable economic growth. AI technologies offer powerful tools for language documentation and can enhance accessibility and usability of Indigenous languages in digital spaces. This study explores both challenges and opportunities associated with leveraging AI to preserve, promote, and utilize indigenous languages as a catalyst for socio-economic development. The application of AI in this context is hindered by several challenges, like the limited representation of indigenous languages in AI datasets, insufficient funding, lack of infrastructure, and biases in AI models trained predominantly on major global languages. Despite these challenges, AI presents opportunities for fostering linguistic inclusivity, promoting multilingual AI applications, and supporting language learning. Through strategic investment in indigenous language digitization, corpus development, and AI-powered translation and speech recognition tools, Zimbabwe can harness its linguistic diversity to foster inclusive and sustainable economic growth. AI can support the creation of tailored language learning tools, translation services, and digital content in indigenous languages to enhance accessibility and cultural preservation. This study employs the qualitative approach which, includes document analysis and semi-structured interviews with linguists, educators, and indigenous community representatives. This study underscores the need for an interdisciplinary approach, where AI solutions are tailored to local linguistic and socio-economic contexts, ensuring that digital advancements contribute to both cultural preservation and economic empowerment. Addressing these challenges through inclusive policies, sustained investment, and community engagement could unlock AI's full capacity to preserve and promote Zimbabwe's rich linguistic diversity. Promoting indigenous language innovation through AI-driven research can

stimulate entrepreneurship, create jobs, and address sector-specific challenges in commerce, industry, agriculture, healthcare, and education.

Key words: Artificial Intelligence (AI), Digital inclusion, Multilingual AI systems, Machine Learning (ML), Digital Language Resources, Language Documentation, Data Scarcity, Digital Divide, Sustainable Economic Growth, Algorithmic Bias, Automated Transcription.



AN IMPROVED LINAC MECHANICAL QUALITY ASSURANCE DEVICE (THE LINATHAD) AND ITS IMPLICATIONS IN THE MEDICAL FIELD.

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The theme of this research is technology enabled healthcare with special focus on cancer treatment using the external beam radiation therapy modality. The electronic device developed here improves both efficiency (by reducing time spend on quality assurance by the Medical Physicist) and accuracy (by providing digital readings to two decimal places). The upkeep (maintenance, calibration and service) of radiotherapy equipment is critical in any centre offering radiotherapy services. This research focuses on the medical linear accelerator machine and the room in which it is housed. The device is an improved version of the previously developed linac mechanical quality assurance verification device. The device (code named the LINATHAD) is microcontroller based and uses modern sensor technologies to sense and quantify physical parameters that are critical in quality assurance of the medical linear accelerator. The device was tested and calibrated with reference to secondary standards in a local government hospital setup in Zimbabwe. Accuracy was tested by comparison between device and calibrated linear accelerator readings and deviations noted. In both cases, three sets of data were generated for each parameter test point and averages were taken as final readings. Other parameters like humidity, temperature and time were calibrated to secondary standards using calibrated discrete devices. Before characterization and calibration, a linear relationship was observed between the device and equipment readouts; data sets had a correlation of 99.96% and a confidence level of 99.98%. Great improvement in both source to surface distance and gantry angle measurement were observed. Device errors were reduced by about Additional

features (humidity and time) also reduced the number of instruments used in mechanical quality assurance of the linear accelerator. The expected contributions of this research to the conference are to serve as a wakeup call to the conference audiences and researchers on how we can improve cancer treatment in Africa using low-cost technology. Collaborations or combined efforts to get solutions to challenges in this sector will support most governments visions and objectives in radiotherapy; most governments have heavily invested in this sector and unfortunately getting less in terms of admirable outputs giving an impression that government efforts are almost failing to give positive outcomes. This contribution can be looked at from various perspectives such as how we could efficiently operate with the few medical physicists in Africa as a whole by increasing productivity, improve cancer treatment (through addressing quality control and quality assurance issues), maximize on the efficient usage of radiotherapy equipment (which are scarce and very expensive). If successfully implemented, all this would consequently lead to improving patient quality of life.

Keywords: Medical Linnear accelerator, Quality Assurance, Gantry angle, Source to Surface Distance, Source to Axis Distance



COLONIALISM'S ENDURING LEGACY: UNPACKING THE ROOTS OF RURAL UNDERDEVELOPMENT, CURVING A PATH FOR GROWING THE RURAL ECONOMY IN ZIMBABWE-THE CASE OF NEMBUDZIYA GOKWE NORTH

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This study argues that colonialism underdeveloped African rural areas through discriminatory policies, and there is need to give rural development a priority. The study undertakes a critical examination of the lasting impacts of colonialism on rural Africa's economic, social, and political trajectories. Through a nuanced historical and comparative analysis, this research reveals the complex mechanisms by which colonial powers systematically underdeveloped rural African economies through urban bias, perpetuating dependence on external markets and distorting local economic systems and leaving rural areas marginalised with pathetic infrastructure. Through the use of desktop research, the study highlights the deliberate disempowerment of rural people and rural economies in general by the white minority, through land grabbing and the displacement of rural communities. It further elucidates the impacts of colonial instruments such as the Hut Tax of 1894,

the Native Reserve Act of 1899, the Native Land Husbandry Act of 1930, the Land Tenure Act of 1969, which favoured whites only, while eroding Indigenous Knowledge Systems and cultures. In addition, this study illuminates the ways in which post-colonial African governments have grappled with the legacy of colonial underdevelopment, often struggling to reverse entrenched patterns of poverty, inequality, and food insecurity. The impacts of urban bias can still be felt today, so many years after colonialism as rural youths migrate to urban areas where there are better facilities, creating yet more problems like pressure on accommodation facilities, burst sewer pipes among other urban problems in that vicious cycle of poverty for Zimbabweans. Views, perceptions and recommendations were sought from 10 key informants and 50 rural residents from Gokwe North district through the use of interviews and focus group discussions. To chart a way forward towards rural transformation, this study proposes a multifaceted framework for rural transformation, encompassing; rural industrialization and agricultural modernization where rural-based industries and improvement of agricultural productivity through technological innovation and capacity building are key. Investing in rural infrastructure and implementing targeted social protection programs to enhance the resilience and well-being of rural households is also crucial to achieve sustainable rural development. There is also need for institutional reforms and policy coordination to support sustainable rural development where policies deliberately favour rural areas considering 67-70% of the population in Zimbabwe resides in the rural areas. By interrogating the historical roots of rural underdevelopment and articulating a comprehensive framework for transformation, this research aims to contribute meaningfully to ongoing debates on sustainable development, poverty reduction, and rural revitalization in Africa.

Keywords: Colonialism, rural development, Rural ecotourism, sustainable transformation, rural industrialization, agricultural modernization.



TECHNOETHICS AND HEURISTIC REASONING: ETHICAL HEURISTICS FOR ADVANCED TECHNOLOGY ADOPTION IN SOUTH AFRICAN DIGITAL TRANSFORMATION INITIATIVES

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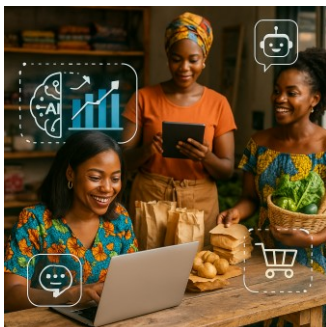
As South Africa accelerates its digital transformation strategies, the ambition of smart city development has gained momentum through national frameworks and strategic policy visions. However, a growing body of scholarship and policy discourse questions the necessity and relevance of smart cities within South Africa and the broader African context. These critiques are grounded in the stark dualities across African societies, where extreme poverty and inequality coexist with sophisticated information and communications infrastructure in select regions. Critics argue that smart cities are neither an immediate priority nor a suitable model in contexts where foundational socio-economic challenges remain unresolved. Against this backdrop, this study proposes a counter-argument. Rather than discarding the smart city vision, it proposes that South Africa should pursue it through the lens of ethical, context-aware technological decision-making. To this end, the paper introduces an *Ethical Heuristics Technological Decision-Making Framework* to guide policy revision on how decision-makers in South Africa should assess, adopt, and implement advanced technologies within smart city initiatives. The framework intends to acknowledge the reality that conditions of uncertainty, a digital divide, limited data, and institutional constraints may lead to heuristic reasoning as a strategic challenge. Moreover, that these challenges may be driven by satisficing and the availability heuristics that are presented as governance limitations, and as impediments of necessary cognitive tools for navigating complex ethical and technological decisions.

Against this backdrop, is the objective of the study that seeks to propose a practical framework beyond technocratic policy models. Furthermore, the study seeks to underpin the proposed framework on the principles of technoethics principles as a lens to address heuristic reasoning during ethical decision-making. These principles are intended to be responsive to the cognitive and contextual realities of advanced digital transformation within South Africa such as those adopted within South African smart cities. Moving from this premise, methodologically, the study will

leverage interdisciplinary literature in digital governance, ethics, and decision science to make a research contribution that is expected to be threefold:

1. The study intends to advance a technoethics heuristic-based model for key technology decisions in advanced technology adoption;
2. The study intends to contribute to deliberations on digital policy ethics, that offers a structured approach to making ethically sound, context-sensitive technology choices.
3. Lastly, the study intends to contribute to the discourse around how decisions should be guided, and to highlight the need for conversations that are centred around ethical decision making during advanced technology planning

The intended presentation of the paper that will be contributed to the conference, intends to invite scholars, policymakers, and practitioners to reconsider the role of human judgment, ethics, and bounded rationality in advanced technology adoption. Furthermore, the presentation of this paper intends to promote and facilitate deliberations that are situated in ethical decision making within the futuristic strategies of advanced and emerging digital technology adoptions.



EXPLORING THE ROLE OF ARTIFICIAL INTELLIGENCE IN EMPOWERING WOMEN-RUNNER ENTREPRENEURS

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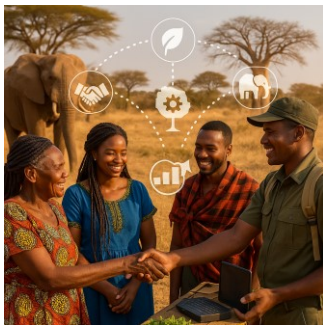
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Artificial intelligence is a powerful tool in this decade that is being utilised in various areas for business enhancement. It promotes entrepreneurship through several processes like bookkeeping and accounts, marketing strategies and product design, etc. AI has been adopted by women entrepreneurs as it helps them improve their sales and reduce their efforts. However, despite its strengths these women are still confronted by many challenges. The present study aims to understand how AI use has impacted migrant women entrepreneurs involved in the runner business in Durban KwaZulu Natal. The investigation was to see how artificial intelligence influences this business, aiming to understand their unique experiences, challenges, and opportunities within the AI technology landscape. This study used a qualitative method, conducting in-depth interviews with ten women entrepreneurs operating businesses within the Durban CBD. The analysis was performed

using thematic analysis, uncovering key themes and insights. The findings reveal that that AI fundamentally affects women entrepreneurs, as they extensively use it to get customers, communicate with customers and retain customers. We infer that where women entrepreneurs embraced AI innovative methods, they managed to get increased profit, it helped them overcome language barriers and provided access to market data. AI was cost effective as it gave them accounting and marketing related services at a less allowing the entrepreneurs to do more with less financial strain. AI also helped in market expansion to other nations. However, in some instances, AI was detrimental to their businesses through the notions of ‘what we ordered vs what we got’ (a trend where the picture of the order and the actual product differ). Some entrepreneurs lacked access to digital tools or training, putting them at a disadvantage in adopting AI technologies. The study emphasises the importance of creating networks for mentorship, guidance, and collaboration to provide women entrepreneurs with platforms to share experiences and resources, enhancing the potential for success in their businesses by leveraging AI.

Keywords: Entrepreneur, Women, Migrant, Artificial Intelligence, Business.



STRENGTHENING COMMUNITY PARTNERSHIP FOR CONSERVATION: GONAREZHOU CONSERVATION TRUST'S ENGAGEMENT WITH SHANGANI PEOPLE IN CHIREDDI, ZIMBABWE

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The Shangani community with its convenience to wildlife conservation faces several challenges such as human-wildlife conflict, livestock predation, limited access to natural resources, economic challenges, habitat encroachment, and land use conflict, climate change impact. These communities in Chiredzi are surrounded by conservation organisations amongst them is Gonarezhou Conservation Trust(GCT), which historically was declared a game sanctuary in 1934 and was completed in 1968, leading to the forced eviction of the Shangani people from their ancestral land. However, the GCT in more recent times, has been making efforts in promoting sustainable wildlife management and community development by involving local communities. This organization integrates corporate governance principles with social responsibilities to address environmental challenges while uplifting local livelihoods. It offers philanthropy, livelihood support climate-resilient agriculture, wildlife management training, and community-based conservation programs. While challenges may

persist particularly regarding land rights and equitable benefit sharing, the integration of corporate governance and social responsibility into conservation strategies has the potential to balance ecological sustainability with socioeconomic empowerment. Therefore, the objective of the study is to assess the corporate governance frameworks adopted by GCT, evaluate its role in conservation in mitigating human-wildlife conflict and preserving cultural heritage, and identify the challenges and opportunities in aligning corporate governance practices with local community needs. The study will employ a mixed methods approach, combining qualitative and quantitative research techniques to provide a comprehensive analysis. Primary data will be collected through surveys which will be conducted within adjacent Shangani communities, interviews, and focus groups with local leaders, conservationists, and park officials on the impact of GCT initiatives on livelihoods, education, and infrastructure. Secondary data will be gathered from policy documents, organizational reports, and historical accounts to analyze the corporate governance frameworks guiding conservation efforts. Data analysis will involve thematic coding for qualitative data and statistical analysis for quantitative responses to identify patterns and correlations. This study is expected to provide insights into the effectiveness of corporate governance initiatives in promoting sustainable development for the Shangani people. It will identify best practices in human-wildlife conflict mitigation and assess the extent to which conservation organisations integrate community needs into their governance structures. The research anticipates highlighting successful models of community engagement and offering recommendations to strengthen collaborative governance. Furthermore, the findings will inform policy decisions and future CSR strategies to ensure equitable benefits for local communities while maintaining biodiversity conservation and community empowerment in Zimbabwe and other similar contexts.

Keywords: Shangani, Gonarezhou Conservation Trust, Corporate Governance and Social responsibility,



THE IMPACT OF RURAL INDUSTRIALISATION ON COMMUNITY HEALTH AND WELL-BEING IN ZIMBABWE.

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This study explores the impact of rural industrialization on community health and well-being, focusing on both the positive and negative effects associated with the growth of industries in rural areas. Zimbabwe is a nation characterized by diverse socio-economic challenges and is heavily reliant on agriculture. As rural areas seek to transition from traditional subsistence farming to industrialized economic activities, this research investigates the multifaceted effects of such transformations on health outcomes and overall well-being. The primary objective is to assess how industrial development influences the physical and mental health of rural populations, including changes in access to healthcare, environmental health risks, and the social dynamics of well-being. Despite the increasing trend of rural industrialization, the literature lacks comprehensive analyses that connect industrial growth directly to specific health outcomes in rural communities. Existing studies often focus on urban areas or generalize rural regions without considering the unique challenges and benefits that industrialization brings to rural settings. This research addresses this gap by employing a mixed-methods approach, combining quantitative analysis of health data from rural communities with qualitative insights gathered from interviews with local residents, healthcare providers, and policymakers. The study investigates a range of indicators, including mortality rates, the prevalence of chronic diseases, mental health issues, and access to healthcare services, while also exploring community-level factors such as social cohesion and economic stability. The methodology includes comparative analysis between rural areas with high industrialization and those with minimal industrial presence, to identify the differential impacts on health and well-being. This research aims to inform policymakers and stakeholders about the critical balance between economic development and public health, emphasizing the need for integrated strategies that promote both community well-being and sustainable industrial growth in Zimbabwe's rural landscapes. Ultimately, this study will also contribute to policymaking by providing evidence on how rural industrialization can be managed to promote sustainable development without compromising community health.

Keywords: Industrialisation, rural, health, well-being.



SUSTAINABLE FINANCING RURAL ECONOMIC DEVELOPMENT IN SOUTHERN AFRICAN COUNTRIES. A SYSTEMATIC REVIEW

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Many rural communities of Less Economically Developed Countries (LEDs) are lagging in terms of economic development. Several challenges have been long experienced by the rural people in Less Economically Developed Countries (LEDs). The most common challenges faced the rural economies of the LEDs include high poverty rate, limited access to economic markets, poorly developed infrastructure, shortage of financial resources. Also, many rural economies of LEDs suffered from backwash effects as well as being a dumping ground for the waste substances from industrial processing. The problem of the study is low levels of economic development in many rural areas of the LEDs. Many rural areas in Africa are the source of economic development in providing mineral, agricultural and human resources. Rural areas are used as a political ground for empty promises during the time of political elections by many politicians in Africa. The main objectives of the research study are; to establish the current state of development in rural areas of LEDs in Africa, to analyse various interventions by the authorities in boosting economic development in rural areas, to examine the impact of sustainable financing of rural communities on the development of the national economy, to assess the challenges for implementing economic development initiatives in rural areas, to establish strategies for sustainable financing of rural economic development. The study is guided by two theoretical frameworks being sustainable development theory and the development finance theory. The scope of the study is on Less Economically developing countries of Africa. The study adopted a systematic review by utilising the existing secondary literature from high impact Scopus journals, and other high rated research papers from conference proceedings.

Key Words: Economic development, developing countries, growth of rural areas, Sustainable financing,



UNPACKING POLICY THEORY AND PRACTICE: EVALUATING SERVICE PROVISION AND UPTAKE IN HIGHER EDUCATION IN ZIMBABWE

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This paper critiques the adequacy and impact of Zimbabwe's ICT policies and legal frameworks on technological advancements and uptake in higher education institutions. It also assesses the response of key stakeholders to the legal frameworks and proposes that they are inadequate, having bred hesitancy. The paper focuses on the analysis of these existing policies and legal instruments to assess their actual impact on higher education institutions (HEIs) and digital transformation. In order to establish a comparative assessment and contextual framework, the literature review examines ICT policies and frameworks adopted by other digital economies both successful and otherwise, highlighting key challenges faced. The study acknowledges the existence of recent legislation, including the Data Protection Act and the Cyber and Data Protection Act, and evaluates its implications, while noting ongoing efforts by the legislator, through the Ministry of Information, Communication Technology, Postal and Courier Services (MICTPCS), to develop an AI policy. The research suggests that with well-structured legal frameworks, Zimbabwe can accelerate its digitalisation efforts, particularly within the higher education sector. The research adopts a mixed methods approach and employs both a desktop study and survey methodology. The desktop study analyses prior research sourced from Google Scholar, Scopus, policy documents, and legislative statutes, employing textual analysis for data interpretation. This is then supported by a survey, using questionnaires to test the findings of the desktop study. Using purposive sampling, respondents will be drawn from officials of selected HTEIs, Ministry of Higher and Tertiary Education Innovation Science and Technology Development, and Ministry of Information, Communication Technology, Postal and Courier Services. The study concludes with recommendations on policy renewal, the integration of ICT policies in higher education, and the engagement of public-private partnerships to support funding, advocacy, and suggestions to assist in the development of a strategic legal framework for Zimbabwe's digital future.

Keywords: Digitalisation, strategic, legal frameworks, integration, ICT policies, public-private partnerships.



SOCIETAL IMPLICATIONS OF TECHNOLOGY, SMART CITIES, AND SMART VILLAGES IN AFRICA

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Africa is experiencing rapid technological growth, influencing various aspects of society, from urbanisation to rural development. The rise of smart cities and smart villages presents both opportunities and challenges, reshaping economies, governance, education, and social structures. The purpose of this study is to extensively review the extant literature on the societal implications of technology, smart cities and smart villages in the African continent, and the challenges of infrastructure gaps, financial barriers and policy support. The study will also recommend ways of accessing financing, to ensure transitioning to a green economy while addressing the negative implications that might arise from turning rural villages and cities into smart ones through technology. This study will employ a systematic literature review analysis. It is anticipated that findings will show that technology in Africa presents vast opportunities for social and economic development, but challenges like inequality, cybersecurity, and policy gaps must be addressed. Policies should be enacted to enhance technology uptake.

Key Words: Africa, Cities, Development, Economy, Green, Smart, Sustainable, Technology, Transition, Villages



SANCTIONS, PROTECTIONISM, AND FOOD; ASSESSING THE CONSEQUENCES FOR AFRICAN AGRICULTURAL TRADE AND FOOD NUTRITIONAL SECURITY

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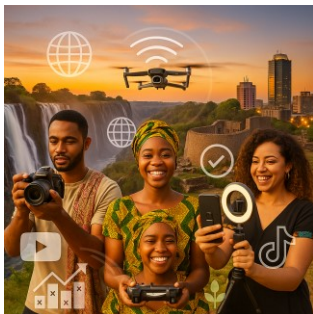
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The Global dynamic in trade and agriculture has shifted since the onset of the Covid-19 pandemic. The complexity of the global trading environment impact and consequences due to the interplay of sanctions, growing protectionism, the weaponization of trade instruments, and the resultant impact on food and nutritional security. Leaders in Africa, should shift their focus towards mitigating and resolving the implications on African agricultural trade and food and nutritional well- being within the context of the evolving global trading and political landscape. Africa's agricultural systems face a multifaceted challenge within the current geopolitical climate, a rise in protectionist sentiments alongside the imposition of sanctions on various nations, geopolitical conflicts and the policies stemming from the Trump administration and the enduring European agricultural subsidies, presents a unique opportunity to analyse the impact of these dynamisms. Taking into context African countries historical experiences, contending with trade restrictions, food crises, the BRICS nations presenting a new influence and alternative to existing trade partners and access to potential developmental investments. This paper examines and assesses the inherent risks of a fragmented and evolving global trading environment as a result of increased market disruptions, geopolitical tensions, ever-increasing food prices and cost of living, the limited access to essential inputs and the forces exacerbating existing challenges and vulnerabilities in Africa's food and nutritional security. Simultaneously, the current situation presents the opportunity to explore the current circumstances and the potential for strengthening South-South cooperation, the drive to boost and promote intra-Africa trade and regional trade integration, to develop a more resilient and self-reliant agricultural systems and value chains. The study utilizes secondary data from global databases, government reports, international trade data (e.g., WTO, FAO, World Bank), and agricultural production and consumption statistics. Descriptive statistics is employed to summarize trends in agricultural production, trade, and food security indicators. Thematic analysis is being used in the study to complement quantitative analysis by offering a deeper, narrative-driven interpretation of the data trade data. Through thematic analysis, recurring patterns, themes, and trends within the data will be

farming in enhancing climate-resilient food systems in Northern Nigeria. This methodological approach allows for the collection of data at some selected regions in Nigeria-Northern region, enabling an analysis of relationships between variables such as innovation adoption, farming outcomes, and climate resilience. This selection is based on the fact that Northern Nigeria is widely recognized as a major agricultural hub, contributing significantly to the nation's food production. This study contributes novel, context-specific evidence to the sparse body of quantitative research on frugal innovation in African food systems. By applying Appropriate Technology Theory, it advances theoretical and practical understandings of how localized, affordable innovations can drive sustainable agricultural transformation in low-income, climate-vulnerable regions. Additionally, findings from this study will draw upon policy frameworks and development programs that will prioritize scaling frugal innovations that is tailored to the socio-economic realities of northern Nigerian micro farmers.

Keywords: Frugal innovation, Micro farming, Climate resilience, Food systems, Northern Nigeria



ZIMBABWE THROUGH THE LENSES OF THREE AFRICAN CONTENT CREATORS

Witness Roy

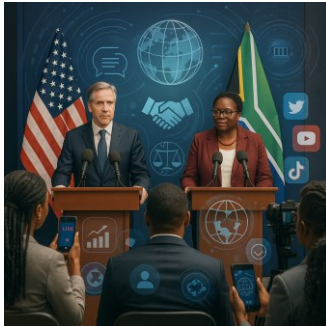
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Ghanaian born Wode Maya, Zinhle Gradwell (South Africa) and Kenyan born African Tigress are three popular African vloggers who have posted generous content about Zimbabwe. Although the trio has painted a positive impression of the country, Zinhle Gradwell is the only one who has been spared by Zimbabwean Facebook users from allegations of being bribed by the Zimbabwean government to disseminate its propaganda. Through participatory culture, this study examines Facebook posts by the three content creators about Zimbabwe with the objectives of (i) identifying techniques which they use (iii) establishing how these techniques contribute towards the portrayal of Zimbabwe (iii) perceptions of Zimbabwean Facebook users towards these representations and responses from the trio. Tentative findings suggest bias in African Tigress' content, a combination of bias and criticism in Wode Maya and a largely apolitical stance on Zinhle Gradwell's posts. It is hoped that this study will, in addition to contributing an exploratory study, inspire tourists to visit Zimbabwe, encourage

critical analysis of the country and urge the government and key stakeholders to work towards improving the lives of Zimbabwean people.

Key words: African vloggers, bias, Zimbabwean Facebook users, content creators



THE US-SA MEDIA BRIEFING AS A THEATRE OF GLOBAL DIPLOMACY IN THE DIGITAL ERA

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Like every other area, diplomacy has been significantly transformed by technological changes. Where meetings between presidents of two countries were held behind closed doors and were bilateral, the interactions can be watched in real time across the world. This calls for soft skills as there is no room for editing. It has opened leaders to scrutiny. However, the presence of cameras may influence some leaders to play to the gallery at the expense of dealing with issues at hand. Trump has been accused of getting (mis/dis)information from his advisors who have South African right-wing roots, the 'Paypal Mafia' led by Elon Musk and conservative Afrikaner formation, Afriforum, resulting in his sensational claims that the South African government is condoning the invasion of white people's properties and their mass killing. In the wake of the first cohort of Afrikaners who sought 'refuge' under Trump's special dispensation, South African president Cyril Ramaphosa led a delegation comprising members of his cabinet, business leaders to the US to "reset" relations between the two countries. The two parties held a media briefing in the Oval Office on the 21st of May 2025 which was beamed across the world where the maverick US president reiterated his allegations which, however received outsize rejection from global news media. White members of the SA delegation refuted Trump's allegations, with businessman Johan Rupert pleading with Musk to empower businesses to combat crime which is indiscriminate. Nonetheless, the SA delegation's trip was met with mixed fortunes by the country's citizens with some hailing Ramaphosa's soft skills for averting a 'Zelensky moment' whilst others accused him of failing to call Trump out for spreading misinformation and disinformation. Ramaphosa not only dismissed rumours that the US had set a condition for SA to amend its telecommunication laws to licence Musk's Starlink but he also said Starlink had never been mentioned in the discussion. Through critical discourse analysis, this paper attempts to provide a thoroughgoing exegesis of the media briefing which featured Trump, the media and the South African

delegation in the Oval Office. This will be done through listening to a recording of the conversation, responses from politicians, political analysts, different South Africans and analysing media coverage around the world. Research objectives include (i) examining the evolution of diplomacy in the Digital Era (ii) the role of technology in international relations and the spread of (dis/mis/information) (iii) assess the extent to which the Oval Office briefing provided a platform for diverse opinions and (iv) perceptions from different groups about the briefing. It is hoped that African diplomats and political leaders leverage technology to develop soft skills to bolster their engagements with other countries. The paper also seeks to urge everyone to verify sources of information.

Key words Trump, Ramaphosa, diplomacy, technology, soft skills, Starlink



GROWING RURAL ECONOMIES LEVERAGING BEEKEEPING: INCOME IMPLICATIONS OF BEEKEEPING VALUE CHAIN INTEGRATION.

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Rural areas present several opportunities that can be used to promote growth of rural economies. Rural activities that can promote the growth of rural economies are widely discussed in literature ranging from on-farm to off-farm activities to include forestry non-timber produce like beekeeping. Largely missing from these discussions is empirical evidence of the income implications of these activities. This study focused on quantifying the income potential of beekeeping in rural areas under different scenarios normally followed by small-scale beekeeping as they migrate from different sequential hurdles of beekeeping adoption. Using a sample of 300 rural households randomly selected from the rural areas of the Eastern Cape province of South Africa the study estimated the income impact of beekeeping participation, commercialization and value addition using the endogenous switching regression model. The study results reveal that while participation in beekeeping has a positive impact on rural household income, beekeepers will significantly increase their income if they commercialize and add value to their honey. The study thus far argues that, if beekeeping is to be targeted to grow rural economies, beekeeping projects promoted in this subsector should focus more on formal honey market linkages and honey value addition promotion.

Key words Beekeeping, honey value addition, commercialization



Strengthening Farmer's Adaptation to Climate Change Impacts on Pest Management, Weather Forecasting and Crop Planning: How Appropriate Are Digital Tools?

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In today's world, climate change is one of the greatest challenges that are increasingly reshaping the earth's ecosystems. Climate change encompasses long-term shifts in temperatures and weather patterns. Human activities such as deforestation, burning of fossil fuels like coal, oil and gas constitute the main drivers of climate change. Owing to greenhouse emissions induced by human activities, the average temperature has increased by 0.9oC since the 19th Century. In line with global climate change, South Africa has experienced an increasing warming trend, with attendant rises in average temperatures and drought conditions and heat across the country. To ensure food security, environmental stability, and economic stability, farmers need to adapt to the vagaries of climate change in managing pest, leveraging weather forecasts, and planning their crops. The study will use hybrid methodologies in the form of surveys and exploratory research. This project will evaluate the adaptation approaches of farmers across the regions of South Africa, by exploring digital tools as the means of strengthening their adaptation to climate change. The study will use a Likert-type scale structured research questionnaire for the data collection. The questionnaire will be designed to elicit responses from the respondents. The researchers will develop a multifactor Climate Change Impact on Farming Practices (CCIFP), adapted from the World Bank's Enterprise Survey and World Bank and International Financial Cooperation Business Environment and Enterprise Surveys indicator descriptions. The overall aim of the study is to provide digital tools-based evidence to strengthen farmers' adaptation to change impacts on pest management, weather forecasting, and cropping planning. This research could inform policies and practices that help ensure a stable food supply in the face of climate change.

Keywords: Climate change, Farmer adaptation, Digital tools, Food security, South Africa



ENTREPRENEURIAL CREATIVITY AND SMALL FIRM INTERNATIONALISATION: A THEORETICAL NEXUS

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This study investigates how entrepreneurial creativity functions as a meta-dynamic capability, crucial for enabling small firm internationalization within resource-constrained environments. Utilizing Uganda as the empirical context, we undertake an in-depth exploration of three exemplary small firms including Endiro Coffee, Kona Uganda, and Delight Uganda Ltd that have successfully penetrated international markets despite significant resource limitations. Anchored theoretically in Dynamic Capabilities Theory, our research employs the Gioia methodology, involving rigorous qualitative analysis of data gathered from 15 semi-structured interviews with firm founders, managers, and stakeholders. Our comprehensive analysis reveals six aggregate dimensions illustrating how entrepreneurial creativity directly facilitates internationalization by fostering opportunity recognition, resource mobilization, and strategic adaptation. Specifically, we find that entrepreneurial creativity in small firms transcends mere product innovation, acting instead as an overarching organizing logic. This logic dynamically activates core processes of sensing market opportunities, seizing strategic resources, and transforming operational frameworks to fit international contexts. This research significantly contributes to international entrepreneurship literature by: (1) conceptualizing creativity as a firm-level meta-capability, distinct from traditional views that restrict it primarily to product or service innovation; (2) enriching dynamic capabilities theory with empirical insights from emerging markets characterized by resource scarcity; and (3) providing practical, actionable insights into how small firms can successfully internationalize through imaginative and relational strategies rather than conventional internationalization pathways. Ultimately, the study underscores entrepreneurial creativity as a vital yet under-explored strategic asset in enabling small firms' successful international market entry.

Keywords: Entrepreneurial creativity, Dynamic capabilities, Small firm internationalization, Resource-constrained environments, Uganda



EXPLORING THE USE OF CENTRAL BANK DIGITAL CURRENCY (CBDC) IN AFRICAN TAX ADMINISTRATION.

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Digitalization has initiated a transformational wave across diverse sectors of the world, not least within finance. The rapid evolution of technological trends, initially sparked by the Third Industrial Revolution and now propelled by the Fourth Industrial Revolution (4IR), prompts a shift from traditional financial paradigms to innovative business models. Within this technological flux, Central Bank Digital Currencies (CBDC) have emerged as a disruptive digital innovation within the financial sector. CBDCs are predicted to revolutionize business operations, tax administration interactions, socioeconomic norms, and government processes, thereby redefining the relationship between the business world and its stakeholders. Digitalizing financial systems at the heart of the 4IR presents unprecedented opportunities and intricate challenges. The RBZ's trial for a CBDC, using distributed ledger technology, aims to facilitate the 'atomic' settlement of payments, provide simultaneous exchanges, and reduce associated risks. CBDCs have been identified as potentially beneficial tools in enhancing tax administration and compliance. They provide tax authorities with greater oversight of transactions, which can mitigate potential tax evasion and improve compliance. However, the utilization of CBDCs within the tax administration landscape is still in its early stages, with many unexplored implications and potential challenges. Key among these is the question of data protection and privacy. Although CBDCs can assist tax administrators in monitoring taxpayer compliance, significant data protection concerns require consideration. Additionally, the introduction of CBDCs has highlighted policy gaps in taxing virtual currencies, indicating the need for comprehensive regulatory frameworks. This research aims to comprehensively analyze the potential benefits, risks, and operational implications associated with using CBDCs in tax administration and compliance, specifically in a Zimbabwean context. This exploration is crucial to inform policy decisions and potential adjustments to existing tax regulatory frameworks in the face of these emerging digital technologies. The study adopted a qualitative research design and use interviews for data collection.

Keywords: Central Bank Digital Currencies, Tax administration, Compliance, Digitalization, Zimbabwe



EFFECTUATION AND BRICOLAGE AND THEIR APPLICABILITY TO SUB-SAHARAN AFRICAN ENTREPRENEURSHIP

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Traditional, Western approaches to entrepreneurship do not adequately capture what is occurring in the sub-Saharan African context. This paper reviews the extent to which specific alternative theoretical perspectives of entrepreneurship are present/applicable in/to sub-Saharan Africa. Over the past two decades, more established economics-based views of entrepreneurial behavior have been supplemented with alternative theoretical approaches to scholarship in the entrepreneurship field; including effectuation and entrepreneurial bricolage. This paper reviews these literatures with a view to what is occurring and applicable in/to the sub-Saharan African environment. There is increasing evidence of effectuation and bricolage in the activities of entrepreneurial actors in resource-constrained environments such as much of sub-Saharan Africa. Emphasizing affordable loss as opposed to expected returns; exploiting networks and social capital through collaboration rather than obsessing over competitive analysis; leveraging contingencies rather than exploiting pre-existing intellectual capital; and adapting to an unpredictable future rather than forecasting an uncertain one, effectuation and bricolage appear to be relevant and exploited entrepreneurial approaches in much of Africa. The aim of this paper is to review two non-traditional/mainstream entrepreneurship approaches in the sub-Saharan African context; given recent research that suggests that effectual decision-making can be utilized beyond the context of uncertain environments (Welter & Kim, 2018) and that entrepreneurial bricolage is regularly employed to overcome resource constraints in sub-Saharan Africa (Ciambotti et al., 2023). Researchers have begun investigating the role that effectuation might play in settings that, despite being innovative, deviate from the more traditional, Western entrepreneur-led venture creation settings in the extant literature. With attention to Africa, this paper supports attempts to increase the generalizability of effectuation to what Futterer et al., (2018, p. 179) call a “theory of decision-making and action under uncertainty”. This is the root of the present paper—non-traditional theories of entrepreneurship in the context of non-traditional entrepreneurship in a geographic context historically under-represented in the literature.

Although the tide is shifting, as a theory of entrepreneurship, there has been a paucity of research on the applicability of entrepreneurial decision-making logics, such as effectuation, to contexts outside of traditional, Western domains such as sub-Saharan Africa.

Keywords: Entrepreneurship, Effectuation, Bricolage, Resource-constrained environments, Sub-Saharan Africa



BRIDGING BORDERS, BUILDING FUTURES: THE ROLE OF INTERNATIONAL EDUCATION IN SHAPING AFRICA'S WORKFORCE AND ENTREPRENEURIAL ECOSYSTEM.

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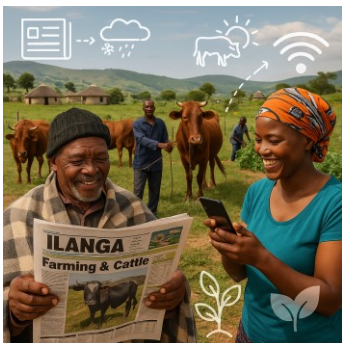
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As Africa approaches a pivotal demographic moment, it is expected to account for more than half of the world's population growth (UN DESA, 2022). The success of the continent's workforce in the future will rely on how well it prepares its youth with global perspectives, entrepreneurial skills, and adaptable skills. In the face of automation, economic informality, and rising youth unemployment, International education presents a potent but underutilized avenue for reshaping Africa's human capital and small business landscape. This paper investigates the role of international education, encompassing study abroad, global virtual exchange, and cross-border academic partnerships in addressing Africa's skills gap and fostering inclusive employment and entrepreneurial ecosystems. The analysis draws from case studies across Ghana, Kenya, Nigeria, and South Africa; semi-structured interviews with alumni of international education programs; and public datasets from UNESCO (2023), the World Bank (2023), and the African Development Bank (AfDB, 2022). In order to evaluate how international academic experiences, affect employability, innovation, and entrepreneurship after graduation, several inputs are combined. According to our research, students who study abroad are far more exposed to high-demand skills including independent thought, digital literacy, problem-solving, and cross-cultural communication. Particularly in high-growth fields like technology, the creative industries, agribusiness, and education, returnee students were more likely to launch startups, run SMEs, or implement international methods in local businesses. However, issues still exist, such as institutional gaps in reintegration support, limited diaspora-inclusive entrepreneurship policies, and misalignment between academic curricula and Africa's rapidly evolving labor market. This research is anchored on the African Union's Agenda 2063 and corresponds with SDGs 4 (Quality Education) and 8 (Decent Work and Economic Growth), emphasizing the need for an ecosystem strategy that connects academic mobility to national development goals. To optimize the local impact

of international exposure, the paper calls for targeted reforms in education policy, university-industry partnership, and frameworks for diaspora engagement. It goes ahead to propose a sustainable development model in which global learning supports local value creation by framing foreign education as a tool for "brain circulation" rather than brain drain. It encourages African higher education institutions, ministries, and international partners to develop inclusive, scalable initiatives that connect global academic capital to local business ecosystems. This study adds a contemporary viewpoint to the discussion on the future of work in Africa by emphasizing foreign education as a means of tackling unemployment, encouraging innovation, and sustaining small company ecosystems. It challenges practitioners, educators, and politicians to investigate how global academic pathways can build a resilient, opportunity-rich future for the continent.

Keywords: International education, Youth employment, Skills development, Entrepreneurship, Africa



EXPLORING THE PROVISION OF CATTLE RELATED AND OTHER AGRICULTURAL INFORMATION USING THE SOUTH AFRICAN ILANGA LOCAL NEWSPAPER IN KWAZULU NATAL PROVINCE OF SOUTH AFRICA

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An exploratory study using a local isiZulu newspaper to disseminate cattle information was undertaken. Communal cattle farmers planned the schedules for Ilanga, a local newspaper. Participatory Action Research (PAR), the cyclic process of plan-act-observe-reflect-plan, formed the basis of this research. One hundred and one (101) articles were published in the Ilanga newspaper over two years. This local newspaper was appropriate due to its large circulation. Articles were published weekly on Thursdays through diversified interactions with stakeholders (including farmers) in the agricultural sector. The feedback from each article was monitored using the number of calls, texts, and WhatsApp messages received. Commodity-based organizations, commercial farmers, smallholders, subsistence or communal farmers, researchers, agricultural advisors or extension officers, animal health technicians, and various research institutions agreed to write article(s) on specific subjects.

Stock theft was used as a case study due to the high number of responses from cattle farmers. However, stock theft was not the subject of all 101 articles. The case study approach allowed an in-depth, multi-faceted exploration of stock theft complexities in cattle farmers' real-life situations. The PAR research approach made it possible to identify issues that affected communal cattle farmers. The study proves that PAR is a multi-dynamic research approach that leads to evolving research situations and unexpected outcomes.

Keywords: Participatory Action Research, Communal cattle farmers, Agricultural communication, Stock theft, isiZulu newspaper



WHEN INNOVATION HARMS INCLUSION: A CRITICAL ANALYSIS OF ANTI-COMPETITIVE RISKS, REGULATORY GAPS, AND SME MARGINALISATION IN SOUTH AFRICA'S EMERGING DRONE ECONOMY

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This paper critically interrogates the paradox at the heart of South Africa's emerging drone economy: while drones are widely promoted as enablers of economic growth, efficiency, and technological advancement, their adoption risks entrenching monopolistic structures and marginalising small and medium enterprises (SMEs). Despite South Africa's pioneering role in establishing drone regulations in 2015, the sector remains underdeveloped, constrained by regulatory inertia, fragmented governance, and uneven market dynamics. Drawing on business ethics and technoethics lenses, the study explores how technological sophistication and interoperability requirements privilege well-resourced incumbents, while compliance burdens disproportionately disadvantage SMEs. The research adopts a qualitative, literature-based scoping methodology, complemented by insights from business leader interviews. Findings reveal two interlinked dynamics: first, the structural advantage afforded to large firms by capital-intensive innovation and interoperability demands; second, the stagnant role of the South African Civil Aviation Authority (SACAA), whose inflexibility exacerbates compliance barriers for new entrants. Although participants acknowledged the competitive benefits of drones, few recognised the ethical responsibility to foster inclusion, framing challenges in narrow market-oriented terms. The analysis highlights a paradox: innovation designed to democratise opportunity may instead reinforce concentration of market power, resulting in exclusionary

practices, market distortion, and reduced innovation diversity. These raises pressing ethical and governance challenges: whether businesses and regulators will act to embed fairness, ensure adaptive regulation, and safeguard SME participation. The paper concludes that inclusive drone innovation requires corporate governance frameworks that prioritise fairness, technoethical approaches to regulation, and sustained stakeholder engagement. Without such measures, South Africa's drone economy risks reproducing the very inequalities that innovation is meant to redress, undermining both competitiveness and inclusive economic growth.

Keywords: Drone economy, small and medium enterprises, Regulatory inertia, Business ethics, South Africa